Insider Disclosures and Stock Returns

INFO 2950 | Phase 2

Group: Vansh Kakra (vk365), Dev Shukla (ds2395), Wilson Wang (ww576)
Mentor: Saif Mahmud (sm2446)

We combine SEC insider disclosures from 2020-2025 with historical S&P 500 membership and stock prices to examine performance after insider purchases and sales. We first prepare the data, then examine the sample and its patterns, and finally specify two analyses for Phase 3.

Language note: Our group does not include a native English speaker.

1. Research Questions

1. Do publicly disclosed insider purchases at S&P 500 companies predict stock returns above those expected from market movements?
2. How does the pattern vary over **1, 4, 13, and 26 weeks**, and how do purchase events compare with sale events?
3. Among purchase events, are **C-suite participation and larger purchases** associated with higher 13-week returns after accounting for the number of insiders?
4. In the later-period sample, do recorded **10b5-1 plans** distinguish purchase events with different outcomes?

We use SEC purchase and sale codes P and S, which can include private transactions as well as exchange trades. We adjust returns for each stock's historical relationship with SPY. The plan question is limited to filings after the checkbox was introduced in 2023.

2. Data Collection

Collecting the Source Files

SEC disclosures: The project obtained the 24 quarterly ZIP archives from **2020q1 through 2025q4** from the SEC Insider Transactions Data Sets. Extracted files are stored in `data/raw/sec/YYYYqQ_form345/`. This notebook reads `SUBMISSION.tsv`, `NONDERIV_TRANS.tsv`, `REPORTINGOWNER.tsv`, and `FOOTNOTES.tsv` from each quarter, preserving a quarter identifier. Other tables in the archives are retained as raw material but are not part of the analysis. The original SEC retrieval date was not recorded.

Index membership: The supplied `data/raw/historical/S&P 500 Historical Components & Changes (Updated).csv` comes from the community-maintained fja05680/sp500 repository. Dated ticker lists allow us to select companies that belonged to the index when filings occurred, including later departures. The original retrieval date is not recorded. The notebook preserves the raw CSV and derives continuous membership intervals.

Prices: `collect_prices.py` downloaded daily prices from Yahoo Finance using yfinance on **October 19, 2026**, requesting data from January 1, 2019 onward for 627 historical index symbols plus SPY. The saved observations span January 2, 2019 through October 19, 2026. The extra pre-2020 history supports market-model estimation, and later prices support follow-up after 2025 filings. The request used `auto_adjust=True` to account for splits and dividends. Failed symbols were retried; unusable results were recorded in `data/raw/prices/daily_stock_prices_missing.csv`. The main file is `data/raw/prices/daily_stock_prices.csv.gz`; the earlier `weekly_stock_prices.csv` remains for reference and is not the Step 3 input. There are 553 stock tickers with data plus SPY and 75 missing symbols.

We use the saved files throughout the notebook. Collection code remains in `collect_prices.py`; cleaning produces separate intermediate and processed files.

Data Sources

We use three sources: **SEC filings** identify insider transactions, **historical S&P 500 membership** determines which companies qualify on each filing date, and **Yahoo Finance prices** measure subsequent stock performance. SPY serves as the market benchmark.

The raw-data folder contains the saved inputs. The collection process below records their coverage and filenames. We describe the final sample after completing the cleaning steps.

Loading the Libraries

We load the Python libraries used to read the source tables, clean and combine records, calculate returns, and produce the charts.

```
# Imports and plot settings
import numpy as np
import pandas as pd
from scipy import stats
import duckdb
import seaborn as sns
```

```
import matplotlib.pyplot as plt
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
```

Loading the SEC Records

We read `SUBMISSION`, `NONDERIV_TRANS`, `REPORTINGOWNER`, and `FOOTNOTES` from each of the **24 quarters in 2020-2025**, combine matching tables, and retain `SOURCE_QUARTER` to trace each record. Fields are initially loaded as text to preserve identifiers and optional fields. The four table types total about **1.31 GB**.

```
# Load the four SEC tables from all 24 quarters, preserving their source quarter.
from pathlib import Path
from time import perf_counter

sec_dir = Path("data/raw/sec")
if not sec_dir.is_dir():
    sec_dir = Path("insider-trading-phase2/data/raw/sec")  # If opened from the repo root.

quarter_folders = [
    sec_dir / f"{year}q{quarter}_form345"
    for year in range(2020, 2026)
    for quarter in range(1, 5)
]
table_names = ["SUBMISSION", "NONDERIV_TRANS", "REPORTINGOWNER", "FOOTNOTES"]
missing_files = [str(folder / f"{table}.tsv")
                 for folder in quarter_folders for table in table_names
                 if not (folder / f"{table}.tsv").is_file()]
if missing_files:
    raise FileNotFoundError("Check the SEC folder location or missing files:\n" + "\n".join(missing_files))

sec_tables = {}
file_counts = []
load_start = perf_counter()
for table in table_names:
    quarter_frames = []
    for folder in quarter_folders:
        frame = pd.read_csv(folder / f"{table}.tsv", sep="\t", dtype="string")
        frame["SOURCE_QUARTER"] = folder.name.split("_")[0]
        file_counts.append({"quarter": frame["SOURCE_QUARTER"].iloc[0] if len(frame) else
folder.name.split("_")[0],
                            "table": table, "rows": len(frame)})
        quarter_frames.append(frame)
    sec_tables[table] = pd.concat(quarter_frames, ignore_index=True)
    del quarter_frames, frame
    print(f"{table}: {len(sec_tables[table]):,} rows loaded", flush=True)

df_submission = sec_tables["SUBMISSION"]
df_transactions = sec_tables["NONDERIV_TRANS"]
df_owners = sec_tables["REPORTINGOWNER"]
df_footnotes = sec_tables["FOOTNOTES"]
df_file_counts = pd.DataFrame(file_counts)
print(f"Loaded {len(file_counts)} files from {len(quarter_folders)} quarters in {perf_counter() -
load_start:.1f} seconds.")
```

```
SUBMISSION: 1,211,888 rows loaded
```

```
NONDERIV_TRANS: 1,841,674 rows loaded

REPORTINGOWNER: 1,323,957 rows loaded

FOOTNOTES: 2,865,764 rows loaded

Loaded 96 files from 24 quarters in 17.1 seconds.
```

Checking the Source Tables

We check the combined table sizes and quarterly row counts for incomplete inputs, then inspect a few transaction rows. These counts include all transaction codes before filtering.

```
# Inspect combined table sizes, source-quarter coverage, and raw transaction fields.
from IPython.display import display

summary = pd.DataFrame([
    {"table": name, "rows": len(frame), "columns": frame.shape[1],
     "quarters": frame["SOURCE_QUARTER"].nunique()}
    for name, frame in sec_tables.items()
])
display(summary)
display(df_file_counts.pivot(index="quarter", columns="table", values="rows"))
print("Transaction columns:", df_transactions.columns.tolist())
display(df_transactions.head())
```

	table	rows	columns	quarters
0	SUBMISSION	1211888	15	24
1	NONDERIV_TRANS	1841674	29	24
2	REPORTINGOWNER	1323957	14	24
3	FOOTNOTES	2865764	4	24

table	FOOTNOTES	NONDERIV_TRANS	REPORTINGOWNER	SUBMISSION
quarter				
2020q1	159716	105852	73693	69408
2020q2	113675	70429	55817	51459
2020q3	92931	60619	43766	39096
2020q4	108077	72973	49045	43607
2021q1	188895	125620	81046	74944
2021q2	140976	89108	66036	58798
2021q3	111270	74002	50774	44057
2021q4	118799	80096	52858	46675
2022q1	171741	110327	76260	71395
2022q2	119120	69774	60772	55261
2022q3	84293	53743	41986	37955
2022q4	89627	58429	42965	38307
2023q1	166835	106093	73681	69457
2023q2	117758	68588	60049	55207
2023q3	85496	53078	41243	37036
2023q4	89079	57852	42436	38108

table	FOOTNOTES	NONDERIV_TRANS	REPORTINGOWNER	SUBMISSION
quarter				
2024q1	167511	111404	71738	67671
2024q2	116519	69412	56838	52802
2024q3	85188	55881	39899	36075
2024q4	94667	63647	43198	38547
2025q1	158489	103029	66287	63284
2025q2	110884	66915	54864	50861
2025q3	85363	55125	39104	35457
2025q4	88855	59678	39602	36421

```
Transaction columns: ['ACCESSION_NUMBER', 'NONDERIV_TRANS_SK', 'SECURITY_TITLE', 'SECURITY_TITLE_FN',
'TRANS_DATE', 'TRANS_DATE_FN', 'DEEMED_EXECUTION_DATE', 'DEEMED_EXECUTION_DATE_FN', 'TRANS_FORM_TYPE',
'TRANS_CODE', 'EQUITY_SWAP_INVOLVED', 'EQUITY_SWAP_TRANS_CD_FN', 'TRANS_TIMELINESS', 'TRANS_TIMELINESS_FN',
'TRANS_SHARES', 'TRANS_SHARES_FN', 'TRANS_PRICEPERSHARE', 'TRANS_PRICEPERSHARE_FN', 'TRANS_ACQUIRED_DISP_CD',
'TRANS_ACQUIRED_DISP_CD_FN', 'SHRS_OWND_FOLWNG_TRANS', 'SHRS_OWND_FOLWNG_TRANS_FN', 'VALU_OWND_FOLWNG_TRANS',
'VALU_OWND_FOLWNG_TRANS_FN', 'DIRECT_INDIRECT_OWNERSHIP', 'DIRECT_INDIRECT_OWNERSHIP_FN', 'NATURE_OF_OWNERSHIP',
'NATURE_OF_OWNERSHIP_FN', 'SOURCE_QUARTER']
```

Table shown with fields as rows for print readability.

	0	1	2	3	4
ACCESSION_NUMBER	0001446472-20-000002	0001084991-20-000014	0001104659-20-041411	0001104659-20-041411	0001104659-20-041409
NONDERIV_TRANS_SK	281943	379212	245379	245380	272457
SECURITY_TITLE	Common Stock	Common Stock	Common Stock	Common Stock	Common Stock
SECURITY_TITLE_FN	<NA>	<NA>	<NA>	<NA>	<NA>
TRANS_DATE	31-MAR-2020	31-MAR-2020	28-MAR-2020	31-MAR-2020	28-MAR-2020
TRANS_DATE_FN	<NA>	<NA>	<NA>	<NA>	<NA>
DEEMED_EXECUTION_DATE	<NA>	<NA>	<NA>	<NA>	<NA>
DEEMED_EXECUTION_DATE_FN	<NA>	<NA>	<NA>	<NA>	<NA>
TRANS_FORM_TYPE	4	4	4	4	4
TRANS_CODE	P	M	F	F	F
...	...	...	...	...	...
TRANS_ACQUIRED_DISP_CD_FN	<NA>	<NA>	<NA>	<NA>	<NA>
SHRS_OWND_FOLWNG_TRANS	1537800.0	8560.0	54890.0	54338.0	24497.0
SHRS_OWND_FOLWNG_TRANS_FN	F2	<NA>	<NA>	<NA>	<NA>
VALU_OWND_FOLWNG_TRANS	<NA>	<NA>	<NA>	<NA>	<NA>
VALU_OWND_FOLWNG_TRANS_FN	<NA>	<NA>	<NA>	<NA>	<NA>
DIRECT_INDIRECT_OWNERSHIP	D	I	D	D	D
DIRECT_INDIRECT_OWNERSHIP_FN	<NA>	<NA>	<NA>	<NA>	<NA>
NATURE_OF_OWNERSHIP	<NA>	By Rabbi Trust	<NA>	<NA>	<NA>

NATURE_OF_OWNERSHIP_FN	<NA>	<NA>	<NA>	<NA>	<NA>
SOURCE_QUARTER	2020q1	2020q1	2020q1	2020q1	2020q1

5 rows × 29 columns

Loading the Membership History

We read the dated ticker lists from the historical membership file. Each row records a membership snapshot. The next steps convert those snapshots into date intervals for matching filings.

```
# Load and inspect the supplied historical membership CSV.
from pathlib import Path

raw_dir = Path("data/raw")
if not raw_dir.is_dir():
    raw_dir = Path("insider-trading-phase2/data/raw")

historical_file = raw_dir / "historical" / "S&P 500 Historical Components & Changes (Updated).csv"
df_historical = pd.read_csv(historical_file)
historical_dates = pd.to_datetime(df_historical["date"], format="%Y-%m-%d")
print(f"Historical membership: {len(df_historical):,} rows, {df_historical.shape[1]} columns")
print(f"Snapshot dates: {historical_dates.min().date()} to {historical_dates.max().date()}")
print("Missing values:", df_historical.isna().sum().to_dict())
display(df_historical.head())
```

```
Historical membership: 2,720 rows, 2 columns
Snapshot dates: 1996-01-02 to 2026-08-18
Missing values: {'date': 0, 'tickers': 0}
```

	date	tickers
0	1996-01-02	AAL,AAMRQ,AAPL,ABI,ABS,ABT,ABX,ACKH,ACV,ADM,AD...
1	1996-01-03	AAL,AAMRQ,AAPL,ABI,ABS,ABT,ABX,ACKH,ACV,ADM,AD...
2	1996-01-04	AAL,AAMRQ,AAPL,ABI,ABS,ABT,ABX,ACKH,ACV,ADM,AD...
3	1996-01-10	AAL,AAMRQ,AAPL,ABI,ABS,ABT,ABX,ACKH,ACV,ADM,AD...
4	1996-01-11	AAL,AAMRQ,AAPL,ABI,ABS,ABT,ABX,ACKH,ACV,ADM,AD...

3. Data Cleaning

Preparing Membership Dates

We parse snapshot dates, standardize ticker spacing and capitalization, and remove repeated tickers and duplicate snapshots. Missing values, empty lists, and conflicting snapshots trigger an error rather than an assumed correction.

Using the SEC filing-date range, we keep the last snapshot before the study begins and all subsequent snapshots through its end. We treat each snapshot date as the start of its recorded membership, subject to the source-date limitation discussed later.

```
# Clean historical membership snapshots and retain coverage for the actual SEC filing window.
filing_dates = pd.to_datetime(df_submission["FILING_DATE"], format="mixed", errors="raise")
if filing_dates.isna().any() or filing_dates.empty:
    raise ValueError("Review missing SEC filing dates before defining the study window.")
study_start = filing_dates.min().normalize()
study_end = filing_dates.max().normalize()
study_end_exclusive = study_end + pd.Timedelta(days=1)

historical = df_historical[["date", "tickers"]].copy()
if historical.isna().any().any():
    raise ValueError("Review missing snapshot dates or ticker lists before cleaning.")
historical["date"] = pd.to_datetime(historical["date"], format="%Y-%m-%d", errors="raise")
ticker_lists = historical["tickers"].astype("string").str.split(",")
ticker_lists = ticker_lists.map(lambda values: [ticker.strip().upper() for ticker in values if
ticker.strip()])
repeated_tickers = int(ticker_lists.map(lambda values: len(values) - len(set(values))).sum())
if ticker_lists.map(len).eq(0).any():
    raise ValueError("An empty membership list requires review.")
historical["tickers"] = ticker_lists.map(lambda values: ",".join(sorted(set(values))))
exact_duplicates = int(historical.duplicated(["date", "tickers"]).sum())
historical = historical.drop_duplicates(["date", "tickers"]).sort_values("date")
if historical["date"].duplicated().any():
    raise ValueError("Conflicting membership snapshots share the same date; review them first.")

baseline = historical.loc[historical["date"].le(study_start)].tail(1)
if baseline.empty:
    raise ValueError("A membership snapshot on or before the first filing is required.")
changes = historical.loc[historical["date"].gt(study_start) & historical["date"].le(study_end)]
historical_snapshots = pd.concat([baseline, changes], ignore_index=True)
historical_snapshots = historical_snapshots.rename(columns={"date": "snapshot_date"})
historical_snapshots["valid_from"] = historical_snapshots["snapshot_date"].clip(lower=study_start)
historical_snapshots["valid_to"] = historical_snapshots["snapshot_date"].shift(-1).fillna(study_end_exclusive)

print(f"SEC filing window: {study_start.date()} through {study_end.date()}")
print(f"Starting membership snapshot: {baseline['date'].iloc[0].date()}")
print(f"Original snapshots: {len(df_historical):,}; retained snapshots: {len(historical_snapshots):,}")
print(f"Exact duplicate snapshots removed: {exact_duplicates}; repeated ticker entries removed:
{repeated_tickers}")
display(historical_snapshots[["snapshot_date", "valid_from", "valid_to"]].head())
```

```
SEC filing window: 2020-01-02 through 2025-12-31
Starting membership snapshot: 2019-12-23
```

```
Original snapshots: 2,720; retained snapshots: 91
Exact duplicate snapshots removed: 0; repeated ticker entries removed: 0
```

	snapshot_date	valid_from	valid_to
0	2019-12-23	2020-01-02	2020-01-28
1	2020-01-28	2020-01-28	2020-03-03
2	2020-03-03	2020-03-03	2020-04-03
3	2020-04-03	2020-04-03	2020-04-06
4	2020-04-06	2020-04-06	2020-05-12

Creating Membership Intervals

We create one row per ticker and membership interval. A filing qualifies when `valid_from <= filing_date < valid_to`. `snapshot_date` preserves the original source date.

We remove incomplete and duplicate ticker/start-date rows, restrict intervals to the study period, shorten overlaps to the next start date, and remove empty intervals.

```
# Expand the retained snapshots into a tidy ticker-by-interval table.
df_historical_clean = historical_snapshots.assign(
    ticker=historical_snapshots["tickers"].str.split(",")
).explode("ticker", ignore_index=True)
df_historical_clean = df_historical_clean[
    ["ticker", "snapshot_date", "valid_from", "valid_to"]
]

# Remove incomplete rows and repeated ticker/start-date pairs.
df_historical_clean = (
    df_historical_clean.dropna()
    .sort_values(["ticker", "valid_from"])
    .drop_duplicates(["ticker", "valid_from"])
    .reset_index(drop=True)
)

# Keep intervals within the study window and end them by the next interval.
df_historical_clean["valid_from"] = df_historical_clean["valid_from"].clip(lower=study_start)
df_historical_clean["valid_to"] = df_historical_clean["valid_to"].clip(upper=study_end_exclusive)
next_start = df_historical_clean.groupby("ticker")["valid_from"].shift(-1)
df_historical_clean["valid_to"] = df_historical_clean["valid_to"].where(
    next_start.isna() | df_historical_clean["valid_to"].le(next_start), next_start
)

# Remove intervals that have no days remaining.
df_historical_clean = df_historical_clean.loc[
    df_historical_clean["valid_from"].lt(df_historical_clean["valid_to"])
].reset_index(drop=True)

print(f"Cleaned membership rows: {len(df_historical_clean):,}")
print(f"Distinct ticker symbols during the study window: {df_historical_clean['ticker'].nunique():,}")
print("Ticker counts are not necessarily unique company counts: symbols and share classes can change.")
display(df_historical_clean.head(10))
```

```
Cleaned membership rows: 45,850
Distinct ticker symbols during the study window: 627
Ticker counts are not necessarily unique company counts: symbols and share classes can change.
```

	ticker	snapshot_date	valid_from	valid_to
0	A	2019-12-23	2020-01-02	2020-01-28
1	A	2020-01-28	2020-01-28	2020-03-03
2	A	2020-03-03	2020-03-03	2020-04-03
3	A	2020-04-03	2020-04-03	2020-04-06
4	A	2020-04-06	2020-04-06	2020-05-12
5	A	2020-05-12	2020-05-12	2020-05-22
6	A	2020-05-22	2020-05-22	2020-06-22
7	A	2020-06-22	2020-06-22	2020-09-18
8	A	2020-09-18	2020-09-18	2020-09-21
9	A	2020-09-21	2020-09-21	2020-10-07

Combining Membership Intervals

We combine consecutive intervals for each ticker while keeping separate periods when a company leaves and later returns. The interval end remains exclusive. We save the result to `data/processed/historical_membership_clean.csv` for matching filings.

```python
# Combine consecutive snapshots into continuous membership periods by ticker.
snapshot_rows = len(df_historical_clean)
previous_end = df_historical_clean.groupby("ticker")["valid_to"].shift()
df_historical_clean["membership_period"] = (
    df_historical_clean["valid_from"].ne(previous_end)
    .groupby(df_historical_clean["ticker"]).cumsum()
)
df_historical_clean = (
    df_historical_clean.groupby(["ticker", "membership_period"], as_index=False)
    .agg(snapshot_date=("snapshot_date", "min"),
         valid_from=("valid_from", "min"),
         valid_to=("valid_to", "max"))
    .drop(columns="membership_period")
)

print(f"Rows before grouping: {snapshot_rows:,}; membership periods: {len(df_historical_clean):,}")
print(f"Distinct ticker symbols: {df_historical_clean['ticker'].nunique():,}")
display(df_historical_clean.head(10))

# Save the cleaned historical membership intervals for reuse.
processed_dir = raw_dir.parent / "processed"
processed_dir.mkdir(parents=True, exist_ok=True)
df_historical_clean.to_csv(processed_dir / "historical_membership_clean.csv", index=False)
print("Saved data/processed/historical_membership_clean.csv")
```

```
Rows before grouping: 45,850; membership periods: 628
Distinct ticker symbols: 627
```

	ticker	snapshot_date	valid_from	valid_to
0	A	2019-12-23	2020-01-02	2026-01-01
1	AAL	2019-12-23	2020-01-02	2024-09-23
2	AAP	2019-12-23	2020-01-02	2023-08-25
3	AAPL	2019-12-23	2020-01-02	2026-01-01
4	ABBV	2019-12-23	2020-01-02	2026-01-01
5	ABC	2019-12-23	2020-01-02	2023-08-30
6	ABMD	2019-12-23	2020-01-02	2022-12-22
7	ABNB	2023-09-18	2023-09-18	2026-01-01
8	ABT	2019-12-23	2020-01-02	2026-01-01
9	ACGL	2022-11-01	2022-11-01	2026-01-01

```
Saved data/processed/historical_membership_clean.csv
```

Loading the Saved Prices

We read the saved daily prices and missing-symbol list. Prices were collected for historical index members and SPY with split and dividend adjustments. Failed symbols were retried, and results with fewer than 20 trading days were treated as unavailable. The output below checks the date range and available tickers before we calculate weekly returns.

```
# Load the saved daily prices and the list of symbols Yahoo could not return.
df_prices_daily = pd.read_csv(raw_dir / "prices" / "daily_stock_prices.csv.gz", parse_dates=["date"])
missing_price_tickers = pd.read_csv(raw_dir / "prices" / "daily_stock_prices_missing.csv")["ticker"].tolist()
print(f"Daily price rows: {len(df_prices_daily):,}; tickers with data: {df_prices_daily['ticker'].nunique()}
(including SPY)")
print(f"Date range: {df_prices_daily['date'].min().date()} to {df_prices_daily['date'].max().date()}")
print(f"Symbols with no Yahoo data ({len(missing_price_tickers)}):", missing_price_tickers)
display(df_prices_daily.head())
```

```
Daily price rows: 1,058,363; tickers with data: 553 (including SPY)
Date range: 2019-01-02 to 2026-10-19
Symbols with no Yahoo data (75): ['ABC', 'ABMD', 'ADS', 'AGN', 'ALXN', 'ANSS', 'ANTM', 'ARNC', 'ATVI', 'AVB',
'BK', 'BLL', 'CDAY', 'CERN', 'CMA', 'COG', 'CTL', 'CTLT', 'CTRA', 'CTXS', 'CXO', 'DAY', 'DFS', 'DISCA', 'DISCK',
'DISH', 'DRE', 'EA', 'EQR', 'ETFC', 'FBHS', 'FI', 'FLIR', 'FLT', 'FRC', 'GPS', 'HBI', 'HES', 'HFC', 'HOLX',
'IPG', 'JNPR', 'JWN', 'K', 'KSU', 'LEG', 'MMC', 'MRO', 'MXIM', 'MYL', 'NBL', 'NLOK', 'NLSN', 'PBCT', 'PEAK',
'PKI', 'PSKY', 'PXD', 'QRVO', 'RE', 'RTN', 'SEE', 'SIVB', 'TIF', 'TWTR', 'UTX', 'VAR', 'VIAC', 'WBA', 'WBD',
'WCG', 'WLTW', 'WRK', 'XEC', 'XLNX']
```

	ticker	date	close	volume
0	A	2019-01-02	62.087963	2113300.0
1	A	2019-01-03	59.800667	5383900.0
2	A	2019-01-04	61.870571	3123700.0
3	A	2019-01-07	63.184349	3235100.0
4	A	2019-01-08	64.110641	1578100.0

Step 1: Cleaning Insider Transactions

We narrow the raw nonderivative transaction table to Form 4 purchase and sale records with usable dates, shares, and prices. Each step reports how many rows remain, preserving the original source tables.

1a. Adding Filing Details

We join transactions to `SUBMISSION` using `ACCESSION_NUMBER`. This adds the filing date, company identifier, ticker, and available plan checkbox. We pad company and owner CIKs to ten digits so formatting differences do not create separate identifiers.

We retain **Form 4 and amended Form 4/A** filings. Form 3 initial holdings and Form 5 annual reports are outside this disclosure-based sample. The filing date anchors the later event-week assignment; it does not provide the exact public-release time.

```
# 1a. Merge filing-level fields onto each transaction, then keep Forms 4 and 4/A.
submission_cols = ["ACCESSION_NUMBER", "FILING_DATE", "DOCUMENT_TYPE", "ISSUERCIK",
                   "ISSUERNAME", "ISSUERTRADINGSYMBOL", "AFF10B5ONE"]
trades = df_transactions.merge(df_submission[submission_cols], on="ACCESSION_NUMBER",
                               how="left", validate="many_to_one")
assert trades["FILING_DATE"].notna().all(), "Every transaction should match exactly one filing."
# Some quarters store CIKs without leading zeros (38777 vs 0000038777); pad so one company has one ID.
trades["ISSUERCIK"] = trades["ISSUERCIK"].str.strip().str.zfill(10)

cleaning_log = [{"step": "Raw non-derivative transactions", "rows": len(trades)}]
trades = trades[trades["DOCUMENT_TYPE"].isin(["4", "4/A"])].copy()
cleaning_log.append({"step": "1a. Forms 4 and 4/A only", "rows": len(trades)})
print(trades["DOCUMENT_TYPE"].value_counts().to_string())
```

```
DOCUMENT_TYPE
4      1793464
4/A      26865
```

1b. Selecting Purchases and Sales

We keep **P (purchase)** and **S (sale)** transaction codes. These include exchange and private transactions. Grants, tax-related share withholding, option exercises, gifts, and other codes are excluded because they describe different types of ownership changes. This keeps the purchase/sale definition consistent throughout the analysis.

```
# 1b. Keep only open-market purchases (P) and sales (S).
print("Transaction codes in Form 4 rows (top 8):")
print(trades["TRANS_CODE"].value_counts().head(8).to_string())
trades = trades[trades["TRANS_CODE"].isin(["P", "S"])].copy()
cleaning_log.append({"step": "1b. Purchase (P) and sale (S) codes only", "rows": len(trades)})
```

```
Transaction codes in Form 4 rows (top 8):
TRANS_CODE
S    546649
A    383521
F    296238
```

```
M    265436
P    164790
J     49083
G     38632
D     36197
```

1c. Checking Dates and Trade Values

We convert transaction and filing dates to dates and shares, prices, and subsequent holdings to numbers. We calculate `trade_value = shares × price` and `filing_lag_days = filing_date - trans_date`, measured in calendar days.

We require positive, observed shares and prices and a filing lag from **0 to 365 days**. Negative lags are inconsistent with the intended transaction-to-filing sequence; the upper limit removes unusually old or potentially miscoded dates. Late filings within that window remain eligible because the analysis follows disclosure. Calendar-day lag alone does not measure compliance with the two-business-day deadline.

```
# 1c. Parse dates and numbers, compute trade value and filing lag, drop unusable rows.
for col in ["TRANS_DATE", "FILING_DATE"]:
    trades[col] = pd.to_datetime(trades[col], format="%d-%b-%Y", errors="coerce")
for col in ["TRANS_SHARES", "TRANS_PRICEPERSHARE", "SHRS_OWND_FOLWNG_TRANS"]:
    trades[col] = pd.to_numeric(trades[col], errors="coerce")

trades["trade_value"] = trades["TRANS_SHARES"] * trades["TRANS_PRICEPERSHARE"]
trades["filing_lag_days"] = (trades["FILING_DATE"] - trades["TRANS_DATE"]).dt.days

valid = (trades["TRANS_DATE"].notna() & trades["FILING_DATE"].notna()
         & trades["TRANS_SHARES"].gt(0) & trades["TRANS_PRICEPERSHARE"].gt(0)
         & trades["filing_lag_days"].between(0, 365))
print(f"Rows dropped (bad shares/price, unparseable dates, or lag outside 0-365 days): {(~valid).sum():,}")
trades = trades[valid].copy()
cleaning_log.append({"step": "1c. Valid dates (lag 0-365 days), shares > 0, price > 0", "rows": len(trades)})
print(trades["filing_lag_days"].describe(percentiles=[.5, .9, .99]).round(1).to_string())
```

```
Rows dropped (bad shares/price, unparseable dates, or lag outside 0-365 days): 9,497

count    700020.0
mean          4.2
std          20.2
min           0.0
50%           2.0
90%           4.0
99%          78.0
max         365.0
```

1d. Resolving Amendments

For each company, owner, transaction date, and transaction code, we check for a Form 4/A amendment. If present, we keep the rows from the latest filing and remove earlier versions. Groups without amendments remain unchanged so separate same-day trades are preserved.

For joint filings, we use the lowest owner CIK as a stable grouping identifier. Step 2 separately uses all reporting owners to assign the filing's highest-ranked role. These rules are reproducible approximations rather than a manual review of every amendment.

```
# 1d. Drop superseded rows when a Form 4/A amends an earlier filing.
first_owner = (df_owners.sort_values("RPTOWNERCIK")
               .drop_duplicates("ACCESSION_NUMBER")[["ACCESSION_NUMBER", "RPTOWNERCIK"]])
trades = trades.merge(first_owner, on="ACCESSION_NUMBER", how="left", validate="many_to_one")
trades["RPTOWNERCIK"] = trades["RPTOWNERCIK"].str.strip().str.zfill(10)

group_key = ["ISSUERCIK", "RPTOWNERCIK", "TRANS_DATE", "TRANS_CODE"]
trades["is_amendment"] = trades["DOCUMENT_TYPE"].eq("4/A")
group_has_amendment = trades.groupby(group_key, dropna=False)["is_amendment"].transform("any")

# Latest filing in each group; ties broken by the larger accession number.
latest_accession = (trades.sort_values(["FILING_DATE", "ACCESSION_NUMBER"])
                    .groupby(group_key, dropna=False)["ACCESSION_NUMBER"].transform("last"))
superseded = group_has_amendment & trades["ACCESSION_NUMBER"].ne(latest_accession)
print(f"Groups containing an amendment: {trades.loc[group_has_amendment,
group_key].drop_duplicates().shape[0]:,}")
print(f"Superseded rows dropped: {superseded.sum():,}")
trades = trades[~superseded].copy()
cleaning_log.append({"step": "1d. Superseded amendment rows removed", "rows": len(trades)})
```

```
Groups containing an amendment: 5,706
Superseded rows dropped: 7,995
```

1e. Saving the Cleaned Transactions

We retain the fields needed for matching and analysis, assign consistent lowercase names, and save `data/interim/sec_ps_trades.csv.gz` . Missing plan flags remain missing. The cleaning log records the change in row counts at each step.

```
# 1e. Select and rename columns, then save the cleaned trades.
df_trades_clean = trades.rename(columns={
    "ACCESSION_NUMBER": "accession", "ISSUERCIK": "issuer_cik", "ISSUERNAME": "issuer_name",
    "ISSUERTRADINGSYMBOL": "ticker_filed", "RPTOWNERCIK": "owner_cik",
    "TRANS_DATE": "trans_date", "FILING_DATE": "filing_date", "TRANS_CODE": "trans_code",
    "TRANS_SHARES": "shares", "TRANS_PRICEPERSHARE": "price", "SHRS_OWND_FOLWNG_TRANS": "shares_after",
    "SECURITY_TITLE": "security_title", "DOCUMENT_TYPE": "form_type", "SOURCE_QUARTER": "source_quarter",
})[["accession", "issuer_cik", "issuer_name", "ticker_filed", "owner_cik", "trans_date",
    "filing_date", "filing_lag_days", "trans_code", "shares", "price", "trade_value",
    "shares_after", "security_title", "form_type", "AFF10B5ONE", "source_quarter"]]

# Ticker cleanup: trim spaces, uppercase, and treat blanks or "NONE" as missing.
df_trades_clean["ticker_filed"] = (df_trades_clean["ticker_filed"].str.strip().str.upper()
                                   .replace({"": pd.NA, "NONE": pd.NA, "N/A": pd.NA}))
df_trades_clean["plan_10b5_1"] = pd.to_numeric(df_trades_clean.pop("AFF10B5ONE"), errors="coerce")
df_trades_clean = df_trades_clean.sort_values(["filing_date", "issuer_cik",
"accession"]).reset_index(drop=True)

interim_dir = raw_dir.parent / "interim"
interim_dir.mkdir(parents=True, exist_ok=True)
df_trades_clean.to_csv(interim_dir / "sec_ps_trades.csv.gz", index=False)
```

```
df_cleaning_log = pd.DataFrame(cleaning_log)
df_cleaning_log["rows_removed"] = -df_cleaning_log["rows"].diff().fillna(0).astype(int)
display(df_cleaning_log)
print(df_trades_clean["trans_code"].value_counts().rename({"P": "purchases", "S": "sales"}).to_string())
print(f"Saved {len(df_trades_clean):,} rows to data/interim/sec_ps_trades.csv.gz")
display(df_trades_clean.head())
```

	step	rows	rows_removed
0	Raw non-derivative transactions	1841674	0
1	1a. Forms 4 and 4/A only	1820329	21345
2	1b. Purchase (P) and sale (S) codes only	711439	1108890
3	1c. Valid dates (lag 0-365 days), shares > 0, ...	700020	11419
4	1d. Superseded amendment rows removed	692025	7995

```
trans_code
sales        537365
purchases    154660
Saved 692,025 rows to data/interim/sec_ps_trades.csv.gz
```

Table shown with fields as rows for print readability.

	0	1	2	3	4
accession	0001127602-20-000197	0000914190-20-000002	0000914190-20-000002	0001271163-20-000001	0001271163-20-000001
issuer_cik	0000001750	0000007623	0000007623	0000009521	0000009521
issuer_name	AAR CORP	ARTS WAY MANUFACTURING CO INC	ARTS WAY MANUFACTURING CO INC	BANCROFT FUND LTD	BANCROFT FUND LTD
ticker_filed	AIR	ARTW	ARTW	BCV	BCV
owner_cik	0001027741	0001022585	0001022585	0001185533	0001185533
trans_date	2019-12-24	2019-12-30	2019-12-31	2019-12-30	2019-12-31
filing_date	2020-01-02	2020-01-02	2020-01-02	2020-01-02	2020-01-02
filing_lag_days	9	3	2	3	2
trans_code	S	P	P	S	S
shares	4793.0	1151.0	51.0	1500.0	2500.0
price	48.02	1.75	1.75	26.02	26.21
trade_value	230159.86	2014.25	89.25	39030.0	65525.0
shares_after	547399.0	2056982.0	2057033.0	150500.0	148000.0
security_title	Common Stock	Common Stock	Common Stock	Common Shares	Common Shares
form_type	4	4	4	4	4
source_quarter	2020q1	2020q1	2020q1	2020q1	2020q1
plan_10b5_1	<NA>	<NA>	<NA>	<NA>	<NA>

Step 2: Adding Roles and Index Membership

We attach insider roles to the cleaned transactions and retain companies that belonged to the S&P 500 on the filing date. This combines the SEC records with the historical membership source.

2a. Assigning Insider Roles

We use the relationship checkboxes and reported titles in `REPORTINGOWNER` to assign roles in the following order:

Rank	Role	Rule
1	C-suite	Officer titled CEO, CFO, COO, or President, excluding Vice President.
2	Other officer	Officer not assigned C-suite rank.
3	Director	Director not assigned an officer rank.
4	10% owner	Large owner not assigned an officer or director rank.
5	Other	Remaining reported relationships.

A filing with multiple owners receives the highest-ranked role among them. Separate director, officer, and large-owner flags retain overlapping relationships. This classification supplies the role variable for the planned analysis; it does not directly measure access to information.

```
# 2a. Build one role record per filing from REPORTINGOWNER.
owners = df_owners[["ACCESSION_NUMBER", "RPTOWNER_RELATIONSHIP", "RPTOWNER_TITLE"]].copy()
relationship = owners["RPTOWNER_RELATIONSHIP"].fillna("")
title = owners["RPTOWNER_TITLE"].fillna("").str.upper()

owners["is_director"] = relationship.str.contains("Director")
owners["is_officer"] = relationship.str.contains("Officer")
owners["is_ten_pct_owner"] = relationship.str.contains("TenPercentOwner")
# C-suite: CEO/CFO/COO by abbreviation or full name, or President that is not Vice President.
names_c_suite = title.str.contains(r"\b(?:CEO|CFO|COO)\b|CHIEF (?:EXECUTIVE|FINANCIAL|OPERATING) OFFICER")
is_president = title.str.contains(r"\bPRESIDENT\b") & ~title.str.contains(r"VICE[\s-]*PRES|\bVP\b")
owners["is_c_suite"] = owners["is_officer"] & (names_c_suite | is_president)

# Rank roles from most senior (1) to least (5); each filing keeps its most senior owner.
owners["role_rank"] = np.select(
    [owners["is_c_suite"], owners["is_officer"], owners["is_director"], owners["is_ten_pct_owner"]],
    [1, 2, 3, 4], default=5)
role_names = {1: "C-suite", 2: "Other officer", 3: "Director", 4: "10% owner", 5: "Other"}
df_roles = (owners.groupby("ACCESSION_NUMBER")
            .agg(role_rank=("role_rank", "min"), is_director=("is_director", "max"),
                 is_officer=("is_officer", "max"), is_ten_pct_owner=("is_ten_pct_owner", "max"),
                 n_owners=("role_rank", "size"), owner_title=("RPTOWNER_TITLE", "first"))
            .reset_index().rename(columns={"ACCESSION_NUMBER": "accession"}))
df_roles["role"] = df_roles["role_rank"].map(role_names)

trades_roles = df_trades_clean.merge(df_roles, on="accession", how="left", validate="many_to_one")
assert trades_roles["role"].notna().all(), "Every filing should have at least one reporting owner."
print(pd.crosstab(trades_roles["role"], trades_roles["trans_code"], margins=True).to_string())
```

```
trans_code          P       S     All
role
10% owner       27225   48856   76081
C-suite         44168  196523  240691
Director        64139  115774  179913
Other            4338    4788    9126
Other officer   14790  171424  186214
All            154660  537365  692025
```

2b. Matching Historical Membership

We match companies to the index **at the filing date**, retaining companies that later left. Ticker spelling and company identifiers can change, so matching uses three steps:

1. Normalize punctuation, spacing, and multiple symbols in filed ticker fields.
2. Link an index ticker to each company CIK accounting for **at least 10%** of its matched filings during membership. This reduces stray matches while allowing company-ID changes.
3. Join trades by CIK and retain filings within the corresponding membership interval.

For simultaneous index share classes, we keep one record per trade and prefer the ticker matching the filing. The output reports linked and unmatched symbols.

```
# 2b. Link index tickers to SEC company IDs (CIKs), then keep trades filed while the company was an S&P 500
member.
import re

def ticker_tokens(raw):
    """Split a filed ticker field into normalized symbols, e.g. 'LEN, LEN.B' -> ['LEN', 'LENB']."""
    if not isinstance(raw, str) or not raw.strip():
        return []
    text = raw.upper().strip()
    if re.fullmatch(r"[A-Z](?:\s[A-Z])+", text):   # 'N O G' -> 'NOG'
        text = text.replace(" ", "")
    parts = re.split(r"[\s,/_;]+|\bAND\b", text)
    return [re.sub(r"[^A-Z]", "", part) for part in parts if re.sub(r"[^A-Z]", "", part)]

# Pass 1: one row per (filing, normalized symbol); index tickers normalized the same way.
filings = trades_roles[["accession", "issuer_cik", "issuer_name", "ticker_filed",
"filing_date"]].drop_duplicates("accession")
filings = filings.assign(symbol=filings["ticker_filed"].map(ticker_tokens)).explode("symbol").dropna(subset=
["symbol"])
membership = df_historical_clean.assign(symbol=df_historical_clean["ticker"].str.replace(r"[^A-Z]", "",
regex=True))

# Pass 2: count filings per (index ticker, CIK) while the ticker was in the index; keep CIKs with >= 10%
share.
candidates = filings.merge(membership[["symbol", "ticker", "valid_from", "valid_to"]], on="symbol")
candidates = candidates[candidates["filing_date"].ge(candidates["valid_from"]) &
candidates["filing_date"].lt(candidates["valid_to"])]
links = (candidates.groupby(["ticker", "issuer_cik"])
         .agg(n_filings=("accession", "nunique"), issuer_name=("issuer_name", "first")).reset_index())
links["share"] = links["n_filings"] / links.groupby("ticker")["n_filings"].transform("sum")
dropped_links = links[links["share"].lt(0.10)]
df_ticker_cik = links[links["share"].ge(0.10)]
print(f"Index tickers linked to a CIK: {df_ticker_cik['ticker'].nunique()} of
{membership['ticker'].nunique()}")
print(f"Stray (ticker, CIK) links dropped for < 10% share: {len(dropped_links)}")
unlinked = sorted(set(membership["ticker"]) - set(df_ticker_cik["ticker"]))
print("Index tickers with no matching insider filings:", unlinked)

# Pass 3: attach membership intervals by CIK and keep trades whose filing date falls inside one.
cik_intervals = df_ticker_cik[["ticker", "issuer_cik"]].merge(membership[["ticker", "valid_from",
"valid_to"]], on="ticker")
sp = trades_roles.merge(cik_intervals.rename(columns={"ticker": "sp_ticker"}), on="issuer_cik", how="inner")
```

```
sp = sp[sp["filing_date"].ge(sp["valid_from"]) & sp["filing_date"].lt(sp["valid_to"])]

# A company with two share classes in the index at once (GOOG and GOOGL) would appear twice; keep one row per
trade,
# preferring the index ticker that matches what the insider filed.
sp["filed_symbol_match"] = [t in ticker_tokens(f) for t, f in zip(sp["sp_ticker"].str.replace(r"[^A-Z]", "",
regex=True), sp["ticker_filed"])]
trade_key = ["accession", "issuer_cik", "owner_cik", "trans_date", "trans_code", "shares", "price",
"security_title"]
sp = (sp.sort_values("filed_symbol_match", ascending=False)
        .drop_duplicates(trade_key).drop(columns=["valid_from", "valid_to", "filed_symbol_match"]))
cleaning_log.append({"step": "2b. Company in S&P 500 on filing date", "rows": len(sp)})
print(sp["trans_code"].value_counts().rename({"P": "purchases", "S": "sales"}).to_string())
```

```
Index tickers linked to a CIK: 611 of 627
Stray (ticker, CIK) links dropped for < 10% share: 8
Index tickers with no matching insider filings: ['AGN', 'AMTM', 'ARES', 'CPRI', 'CRH', 'CVNA', 'ERIE', 'FIX',
'FOXA', 'FRC', 'NWSA', 'PSKY', 'SBNY', 'SOLS', 'VNT', 'WCG']

trans_code
sales        121168
purchases      6361
```

2c. Removing Duplicate Trades

We remove repeated records sharing the owner, transaction date, code, shares, price, and index ticker, including trades reported under related company CIKs. We then save the matched trades to `data/interim/sp500_trades.csv.gz` and the ticker links to `data/interim/sp500_ticker_cik_links.csv` .

```
# 2c. Drop cross-CIK duplicates, then save the S&P 500 trade file.
dup_across_ciks = sp.duplicated(["owner_cik", "trans_date", "trans_code", "shares", "price", "sp_ticker"])
print(f"Duplicate trades reported under a second CIK: {dup_across_ciks.sum():,}")
df_sp_trades = sp[~dup_across_ciks].sort_values(["filing_date", "sp_ticker",
"accession"]).reset_index(drop=True)
cleaning_log.append({"step": "2c. Same trade under two CIKs removed", "rows": len(df_sp_trades)})

df_sp_trades.to_csv(interim_dir / "sp500_trades.csv.gz", index=False)
df_ticker_cik.to_csv(interim_dir / "sp500_ticker_cik_links.csv", index=False)
print(f"Saved {len(df_sp_trades):,} trades in {df_sp_trades['issuer_cik'].nunique()} companies to
data/interim/sp500_trades.csv.gz")
print(pd.crosstab(df_sp_trades["role"], df_sp_trades["trans_code"], margins=True).to_string())
```

```
Duplicate trades reported under a second CIK: 161

Saved 127,368 trades in 593 companies to data/interim/sp500_trades.csv.gz
trans_code         P       S     All
role
10% owner        742    3291    4033
C-suite         1205   51938   53143
Director        3633   23605   27238
Other             20     542     562
Other officer    758   41634   42392
All             6358  121010  127368
```

Step 3: Preparing Weekly Returns

We check daily prices, calculate Friday-to-Friday returns for each stock and SPY, and select one available price history per company. This supplies the return data needed for each disclosure event.

3a. Calculating Weekly Returns

We remove repeated ticker-days and missing or nonpositive closes. We then take the last available close in each Friday-labeled week and calculate **current close / previous close - 1**. A Friday holiday can therefore use Thursday's close. We apply the same calculation to SPY and join its weekly return to the stock returns.

```
# 3a. Clean daily prices, then compute Friday-to-Friday weekly returns for every ticker and for SPY.
prices = df_prices_daily.drop_duplicates(["ticker", "date"])
bad_prices = prices["close"].le(0) | prices["close"].isna()
print(f"Duplicate ticker-days removed: {len(df_prices_daily) - len(prices):,}; non-positive or missing closes
removed: {bad_prices.sum():,}")
prices = prices[~bad_prices]

# Last close of each week ending Friday ('W-FRI'), then the percentage change from the prior week.
weekly_close = (prices.set_index("date").groupby("ticker")["close"]
                .resample("W-FRI").last().dropna().reset_index())
weekly_close["ret"] = weekly_close.groupby("ticker")["close"].pct_change()
df_weekly = weekly_close.dropna(subset=["ret"]).rename(columns={"date": "week"})

market = df_weekly.loc[df_weekly["ticker"].eq("SPY"), ["week", "ret"]].rename(columns={"ret": "mkt_ret"})
df_weekly = df_weekly[df_weekly["ticker"].ne("SPY")].merge(market, on="week", how="inner")
print(f"Weekly return rows: {len(df_weekly):,} across {df_weekly['ticker'].nunique()} stocks; "
      f"{market['week'].min().date()} to {market['week'].max().date()}")
print("Weekly stock returns, percent:")
print((df_weekly["ret"] * 100).describe(percentiles=[.01, .5, .99]).round(2).to_string())
```

```
Duplicate ticker-days removed: 0; non-positive or missing closes removed: 0

Weekly return rows: 219,378 across 552 stocks; 2019-01-11 to 2026-10-23
Weekly stock returns, percent:
count    219378.00
mean          0.34
std           5.21
min         -61.01
1%          -13.87
50%           0.32
99%          15.20
max         129.20
```

3b. Linking Company Price Histories

For each company CIK, we select the linked ticker with available prices and the latest membership end date. This allows older symbols to use a newer linked price history, such as FB and META. We report companies without an available history because their events cannot receive return estimates.

```
# 3b. For each company (CIK), pick the linked index ticker with price data and the latest membership end date.
tickers_with_prices = set(df_weekly["ticker"])
latest_end = df_historical_clean.groupby("ticker")["valid_to"].max().rename("latest_end").reset_index()
price_choice = (df_ticker_cik[["issuer_cik", "ticker"]].merge(latest_end, on="ticker")
                .loc[lambda d: d["ticker"].isin(tickers_with_prices)]
                .sort_values("latest_end").drop_duplicates("issuer_cik", keep="last")
                .rename(columns={"ticker": "price_ticker"})[["issuer_cik", "price_ticker"]])

renamed = (df_sp_trades[["issuer_cik", "sp_ticker"]].drop_duplicates()
           .merge(price_choice, on="issuer_cik").query("sp_ticker != price_ticker"))
print(f"Companies with a usable price series: {price_choice['issuer_cik'].nunique()} of
{df_sp_trades['issuer_cik'].nunique()}")
print("Examples where an old index ticker uses its newer ticker's prices:")
print(renamed.drop_duplicates("sp_ticker").head(10).to_string(index=False))

no_prices = df_sp_trades[~df_sp_trades["issuer_cik"].isin(price_choice["issuer_cik"])]
print(f"\nTrades in companies with no price series: {len(no_prices):,} "
      f"({no_prices['trans_code'].eq('P').sum():,} purchases) in {no_prices['issuer_cik'].nunique()}
companies, e.g.",
      sorted(no_prices["sp_ticker"].unique())[:12])
```

```
Companies with a usable price series: 540 of 593
Examples where an old index ticker uses its newer ticker's prices:
issuer_cik sp_ticker price_ticker
0001326801        FB         META
0001140859       ABC          COR
0000009389       BLL         BALL
0001156039      ANTM          ELV
0001095073        RE           EG
0000031791       PKI         RVTY
0000101829       UTX          RTX
0001175454       FLT         CPAY
0000765880      PEAK          DOC
0000813828      VIAC         PARA

Trades in companies with no price series: 4,511 (271 purchases) in 53 companies, e.g. ['ABMD', 'ADS', 'ALXN',
'ANSS', 'ATVI', 'AVB', 'BK', 'CDAY', 'CERN', 'CMA', 'COG', 'CTLT']
```

Step 4: Building the Event Dataset

We combine trades into **one row per company, event week, and purchase/sale direction**. We then estimate expected stock returns from the market and calculate the difference between actual and expected returns after each event.

4a. Combining Trades into Events

Grouping trades prevents several transactions at one company in the same week from being counted as separate signals. We assign each filing to the first Friday on or after **filing date plus one weekday** and use that week's close as the entry price. This is an assumed timing rule; it does not account for exact filing times or market holidays.

For each event, we count trades and distinct retained owners, sum purchase or sale dollars, and take the natural log of that total. We also retain the highest-ranked role, whether any trade has C-suite rank, the maximum observed plan flag, average filing lag, and whether the company has both purchase and sale

events that week.

```
# 4a. Assign each trade to its event week and aggregate to one row per company, week, and direction.
df_sp_trades["public_date"] = df_sp_trades["filing_date"] + pd.offsets.BDay(1)
# First Friday on or after the public date (rollforward leaves a date that is already a Friday unchanged).
df_sp_trades["event_week"] = df_sp_trades["public_date"].map(pd.offsets.Week(weekday=4).rollforward)

df_events = (df_sp_trades.groupby(["issuer_cik", "event_week", "trans_code"])
             .agg(issuer_name=("issuer_name", "first"), sp_ticker=("sp_ticker", "first"),
                  n_trades=("accession", "size"), n_insiders=("owner_cik", "nunique"),
                  total_value=("trade_value", "sum"), role_rank=("role_rank", "min"),
                  any_c_suite=("role_rank", lambda r: int((r == 1).any())),
                  any_plan_10b5_1=("plan_10b5_1", "max"), mean_filing_lag=("filing_lag_days", "mean"),
                  first_filing=("filing_date", "min"))
             .reset_index())
df_events["role"] = df_events["role_rank"].map(role_names)
df_events["log_value"] = np.log(df_events["total_value"])
both = df_events.groupby(["issuer_cik", "event_week"])["trans_code"].transform("nunique")
df_events["mixed_week"] = both.gt(1).astype(int)
df_events = df_events.merge(price_choice, on="issuer_cik", how="left")
print(df_events.groupby("trans_code").agg(events=("issuer_cik", "size"), companies=("issuer_cik", "nunique"),
                                          trades=("n_trades", "sum")).rename(index={"P": "purchases", "S":
"sales"}).to_string())
```

```
            events  companies  trades
trans_code
purchases     2356        459    6358
sales        25464        582  121010
```

4b. Estimating Expected Returns

We estimate each stock's relationship with SPY using up to **52 available weekly returns before the event**, requiring at least **40 observations**:

```
stock return = alpha + beta × SPY return + residual
```

Here, alpha is the fitted intercept and beta is the stock's response to market movements. Using only earlier returns avoids fitting the benchmark to the outcome being measured. We calculate the single-variable regression slope as covariance divided by market-return variance, then calculate its intercept from the two means.

```
# 4b. Estimate alpha and beta for each event from the 52 weeks before its event week.
weekly_by_ticker = {t: g.set_index("week")[["ret", "mkt_ret"]].sort_index() for t, g in
df_weekly.groupby("ticker")}
ESTIMATION_WEEKS, MIN_WEEKS = 52, 40

def market_model(ticker, event_week):
    """Return (alpha, beta, weeks used) from the 52 weeks strictly before event_week."""
    series = weekly_by_ticker.get(ticker)
    if series is None:
        return np.nan, np.nan, 0
    window = series.loc[:event_week - pd.Timedelta(days=1)].tail(ESTIMATION_WEEKS)
    if len(window) < MIN_WEEKS:
        return np.nan, np.nan, len(window)
```

```
    beta = np.cov(window["ret"], window["mkt_ret"])[0, 1] / window["mkt_ret"].var()
    alpha = window["ret"].mean() - beta * window["mkt_ret"].mean()
    return alpha, beta, len(window)

fits = [market_model(t, w) for t, w in zip(df_events["price_ticker"], df_events["event_week"])]
df_events[["alpha", "beta", "estimation_weeks"]] = pd.DataFrame(fits, index=df_events.index)
print(f"Events with an estimated beta: {df_events['beta'].notna().sum():,} of {len(df_events):,}")
print(df_events["beta"].describe(percentiles=[.05, .5, .95]).round(2).to_string())
```

```
Events with an estimated beta: 26,100 of 27,820
count    26100.00
mean         1.02
std          0.50
min         -1.29
5%           0.25
50%          1.00
95%          1.89
max          4.75
```

4c. Calculating Abnormal Returns

For each week after entry, we calculate **abnormal return = actual return - (alpha + beta × SPY return)**. We sum these differences over **1, 4, 13, and 26 weeks** to obtain cumulative abnormal returns (CARs). Values remain missing when the required price or estimation history is unavailable. Stored returns are decimals; multiplying by 100 expresses them in percentage points.

We also calculate `car_pre4` over the four weeks ending with the event week to examine recent price movement. Because this includes the event week, it can include a response to the disclosure and is not strictly a pre-disclosure measure.

```
# 4c. Abnormal returns: sum (actual - expected) over 1, 4, 13, 26 weeks after the event week, plus the 4 weeks
ending with it.
HORIZONS = [1, 4, 13, 26]

def cumulative_abnormal_returns(ticker, event_week, alpha, beta):
    series = weekly_by_ticker.get(ticker)
    out = {f"car_{h}w": np.nan for h in HORIZONS} | {"car_pre4": np.nan}
    if series is None or np.isnan(beta):
        return out
    abnormal = series["ret"] - (alpha + beta * series["mkt_ret"])
    after = abnormal.loc[event_week + pd.Timedelta(days=1):]
    for h in HORIZONS:
        window = after.iloc[:h]
        # Require h consecutive weekly returns that end within h weeks plus a holiday buffer of the event
week.
        if len(window) == h and (window.index[-1] - event_week).days <= 7 * h + 7:
            out[f"car_{h}w"] = window.sum()
    before = abnormal.loc[:event_week].tail(4)
    if len(before) == 4:
        out["car_pre4"] = before.sum()
    return out

cars = [cumulative_abnormal_returns(t, w, a, b) for t, w, a, b in
        zip(df_events["price_ticker"], df_events["event_week"], df_events["alpha"], df_events["beta"])]
df_events = pd.concat([df_events, pd.DataFrame(cars, index=df_events.index)], axis=1)
```

```
car_cols = [f"car_{h}w" for h in HORIZONS]
print("Events with each CAR available:")
print(df_events.groupby("trans_code")[car_cols + ["car_pre4"]].count().rename(index={"P": "purchases", "S":
"sales"}).to_string())
```

```
Events with each CAR available:
            car_1w  car_4w  car_13w  car_26w  car_pre4
trans_code
purchases     2227    2227     2227     2227      2227
sales        23873   23873    23873    23873     23873
```

4d. Saving the Event Dataset

We order the columns and save `data/processed/insider_events.csv` . Events with missing prices or beta remain in the file for coverage counts, but cannot enter analyses requiring returns. The output shows the complete cleaning log and a preview of purchase events.

```
# 4d. Order columns, save the analysis-ready event table, and print the full cleaning funnel.
event_cols = ["issuer_cik", "issuer_name", "sp_ticker", "price_ticker", "trans_code", "event_week",
"first_filing",
              "n_trades", "n_insiders", "total_value", "log_value", "role", "role_rank", "any_c_suite",
              "any_plan_10b5_1", "mixed_week", "mean_filing_lag", "alpha", "beta", "estimation_weeks",
              "car_pre4"] + car_cols
df_events = df_events[event_cols].sort_values(["event_week", "issuer_cik",
"trans_code"]).reset_index(drop=True)
df_events.to_csv(processed_dir / "insider_events.csv", index=False)

has_returns = df_events["beta"].notna()
cleaning_log.append({"step": "4a. Trades rolled up into company-week events (rows = events)", "rows":
len(df_events)})
cleaning_log.append({"step": "4b. Events with a price series and >= 40 weeks for beta", "rows":
int(has_returns.sum())})
df_cleaning_log = pd.DataFrame(cleaning_log)
display(df_cleaning_log)
print(f"Saved {len(df_events):,} events ({df_events['trans_code'].eq('P').sum():,} purchase events, "
      f"{df_events['trans_code'].eq('S').sum():,} sale events) to data/processed/insider_events.csv")
display(df_events[df_events["trans_code"].eq("P")].head())
```

	step	rows
0	Raw non-derivative transactions	1841674
1	1a. Forms 4 and 4/A only	1820329
2	1b. Purchase (P) and sale (S) codes only	711439
3	1c. Valid dates (lag 0-365 days), shares > 0, ...	700020
4	1d. Superseded amendment rows removed	692025
5	2b. Company in S&P 500 on filing date	127529
6	2c. Same trade under two CIKs removed	127368
7	4a. Trades rolled up into company-week events ...	27820
8	4b. Events with a price series and >= 40 weeks...	26100

```
Saved 27,820 events (2,356 purchase events, 25,464 sale events) to data/processed/insider_events.csv
```

Table shown with fields as rows for print readability.

	1	5	21	27	55
issuer_cik	0000051253	0000940944	0000023217	0000051253	0000822416
issuer_name	INTERNATIONAL FLAVORS & FRAGRANCES INC	DARDEN RESTAURANTS INC	CONAGRA BRANDS INC.	INTERNATIONAL FLAVORS & FRAGRANCES INC	PULTEGROUP INC/MI/
sp_ticker	IFF	DRI	CAG	IFF	PHM
price_ticker	IFF	DRI	CAG	IFF	PHM
trans_code	P	P	P	P	P
event_week	2020-01-03	2020-01-03	2020-01-10	2020-01-10	2020-01-10
first_filing	2020-01-02	2020-01-02	2020-01-03	2020-01-03	2020-01-07
n_trades	2	1	2	1	1
n_insiders	1	1	1	1	1
total_value	8272210.0	163110.0	1359489.0	8729840.0	14909.034
...	...	...	...	...	...
mixed_week	0	0	1	0	0
mean_filing_lag	3.0	2.0	2.0	3.0	4.0
alpha	-0.005327	-0.001759	0.003111	-0.005349	0.006139
beta	1.019928	0.746388	1.340501	1.020515	0.285479
estimation_weeks	51	51	52	52	52
car_pre4	-0.112750	-0.075855	0.086221	-0.061284	-0.039494
car_1w	-0.007962	0.021369	0.018333	0.056359	0.033403
car_4w	0.062931	0.052300	-0.027635	0.110913	0.114467
car_13w	0.090631	-0.443717	0.173048	0.268824	-0.206934
car_26w	0.209370	-0.010890	0.069355	0.192880	-0.000015

5 rows × 25 columns

Saved Datasets

File	Contents
`data/interim/sec_ps_trades.csv.gz`	Cleaned SEC purchase and sale transactions.
`data/interim/sp500_trades.csv.gz`	Transactions matched to historical S&P 500 membership.
`data/interim/sp500_ticker_cik_links.csv`	Ticker-to-company links.
`data/processed/historical_membership_clean.csv`	Cleaned membership intervals.
`data/processed/insider_events.csv`	Final company-week-direction events.

When reloading CSV files, preserve CIKs as strings and parse date columns explicitly.

4. Final Dataset

Purpose

This dataset was assembled for the Cornell INFO 2950 group project by Vansh Kakra, Dev Shukla, and Wilson Wang. Its purpose is to examine whether public insider disclosures contain information associated with subsequent stock returns beyond market movements. This relates to **market efficiency**, the extent to which prices already reflect available information. It is intended for educational, observational analysis, not individual surveillance, investment recommendations, or detecting illegal trading.

This description follows Gebru et al. (2018), *Datasheets for Datasets*.

Sample and Coverage

The analysis file, `data/processed/insider_events.csv`, contains **27,820 rows and 25 columns**. One observation is a unique **(issuer_cik, event_week, trans_code)** combination: one company, Friday-labeled week, and direction. A company with both buying and selling has two rows that week. The 593 companies are not 593 simultaneous index constituents; membership changes across the six-year filing period.

There are **2,356 purchase events in 459 companies** and **25,464 sale events in 582 companies**, derived from **6,358 purchase and 121,010 sale trade records**. Filing dates cover January 2, 2020 through December 31, 2025, while event weeks cover January 3, 2020 through January 2, 2026. The entry rule shifts some late-2025 filings into 2026. Selection is based on filings during the study period, not exclusively transaction dates during that period.

Missing data: 1,665 events lack a linked price ticker. Another 55 have a price ticker but no estimated beta, for **1,720 events without returns** in total. This leaves **26,100 events**, including 2,227 purchases and 23,873 sales, with all four CAR horizons. The 129 purchase events without returns comprise 114 without a price ticker and 15 additional events without beta. No extra 26-week attrition occurs in this saved snapshot. The plan flag is missing for **17,060 events**; missing is not coded as zero. Return exclusions are applied only when a particular analysis requires those values.

The interim files remain at a different unit: `sec_ps_trades.csv.gz` contains 692,025 cleaned SEC P/S trade rows across companies; `sp500_trades.csv.gz` contains the 127,368 matched S&P 500 trade rows. The membership and ticker-CIK files document matching. No train/test split is stored in the event file, and there are no hand-labeled outcomes: CARs are computed quantities.

Event Variables

CSV has no enforced schema. Read CIKs as strings to retain leading zeros and parse date columns explicitly. Returns are stored as decimals; multiply by 100 to express percentage points. The following types describe intended values, including missing numeric values where stated.

Column	Type	Meaning and units
`issuer_cik`	String	Ten-digit SEC company identifier, including leading zeros.
`issuer_name`	String	First retained company name within the event; names may change over time.
`sp_ticker`	String	First retained historical index ticker for the event.

Column	Type	Meaning and units
price_ticker	String, nullable	Linked ticker used for prices, selected by latest membership end among available histories.
trans_code	Category	P for purchase or S for sale, as reported in Form 4.
event_week	Date	Friday on or after filing date plus one weekday; event aggregation and entry label.
first_filing	Date	Earliest filing date among trades in this event.
n_trades	Integer	Number of retained trade rows in the event, not number of filings.
n_insiders	Integer	Number of distinct retained owner CIKs; joint filings can limit interpretation.
total_value	Float	Sum of reported shares times price per share, in nominal dollars.
log_value	Float	Natural logarithm of total dollar value, dimensionless.
role	Category	Most senior assigned role under the project's ranking.
role_rank	Integer	1 C-suite, 2 Other officer, 3 Director, 4 10% owner, 5 Other.
any_c_suite	Binary integer	1 if any retained trade has assigned C-suite rank, otherwise 0.
any_plan_10b5_1	Nullable binary	Maximum observed filing plan flag within the event; missing if all flags are missing. A zero can coexist with missing component flags.
mixed_week	Binary integer	1 if the same company has both P and S events that week.
mean_filing_lag	Float	Mean filing date minus transaction date over event trades, in calendar days.
alpha	Nullable float	Market-model intercept, in weekly decimal-return units.
beta	Nullable float	Market-model slope against SPY, unitless.
estimation_weeks	Integer	Number of available return rows considered, up to 52; 0 when no series exists, with at least 40 required for a fit.
car_pre4	Nullable float	Sum of four abnormal weekly returns ending with the event week; decimal units, not strictly pre-disclosure.
car_1w	Nullable float	Sum of abnormal returns over the first week after the event week, decimal units.
car_4w	Nullable float	Sum over the first 4 weeks after the event week, decimal units.
car_13w	Nullable float	Sum over the first 13 weeks after the event week, decimal units.
car_26w	Nullable float	Sum over the first 26 weeks after the event week, decimal units.

Source Coverage and People

The **SEC** is the primary source of legally mandated disclosures. Its records document reported transactions rather than verified motives or illegal conduct. **Historical membership** comes from a community-maintained resource that adds coverage of departed firms. **Yahoo Finance** supplies adjusted price histories, accessed through the third-party yfinance package. Each source supplies information the others lack: transactions, historical eligibility, and market outcomes, respectively.

Section 16 reporting obligations cover officers, directors, and beneficial owners above the relevant 10% threshold, not every employee or investor. Form 4 changes in ownership generally have a two-business-day reporting deadline. This creates a selected, self-reported sample with varying disclosure delays. See the SEC investor bulletin. The S&P 500 restriction further selects large companies. Vendor coverage, ticker changes, filing corrections, and title wording influence which observations survive matching.

The plan checkbox became required for applicable reports filed on or after **April 1, 2023**, according to the SEC's 10b5-1 disclosure announcement. Earlier missingness cannot be interpreted as absence of a plan. Roles are derived from self-reported relationship fields and titles, not independently verified employment histories.

People and awareness: Filers provide identifying information for a public regulatory disclosure process. Public filing does not establish that they consented to this particular research use. We did not contact insiders or collect research consent, and cannot verify each person's understanding of downstream reuse. Raw files identify individuals and may include addresses; the final event table uses company identifiers and aggregated roles rather than insider names. Owner identifiers in interim data can still be linked to individuals. We avoid individual rankings or allegations.

Preparation

The preceding cleaning steps use pandas and NumPy to combine filings, membership, and prices. yfinance was used for the separate price download.

Steps 1-4 document the selection rules, amendment handling, role assignment, historical membership matching, and price calculations. The final table combines trades by company, week, and direction and attaches the market-adjusted returns. Intermediate files preserve the trade-level records used to construct it.

Access and Intended Use

Suitable uses include descriptive summaries and the limited course-method association models specified below. Unsuitable uses include inferring illegal trading, identifying individual wrongdoing, or presenting a guaranteed profitable strategy. The data do not contain experimental assignments or controls sufficient for causal claims. They are a fixed course-project snapshot, with no promised update schedule or independent audit of all source records.

Reviewers can obtain the saved raw files, intermediate files, and processed files from the project branch. Original sources are the SEC quarterly datasets, fja05680/sp500, and Yahoo Finance through yfinance. We credit the regulator, community maintainers, price provider, and software contributors; inclusion in this educational repository does not imply ownership of their data or unrestricted redistribution rights.

5. Exploratory Data Analysis

5.1 Sample Coverage

We first check the number of trades, events, and companies, followed by typical purchase values and return availability. Missing outcomes remain in coverage counts but are excluded from numerical return summaries. Plan-flag percentages use observed flags among events first filed in 2023 or later; missing flags are counted separately.

```
# 5a. Describe sample coverage, purchase-event predictors, and recorded plan flags.
purchase_events = df_events.loc[df_events["trans_code"].eq("P")].copy()
sale_events = df_events.loc[df_events["trans_code"].eq("S")].copy()
car_cols = [f"car_{h}w" for h in [1, 4, 13, 26]]
sample_rows = []
for direction, label in [("P", "Purchases"), ("S", "Sales")]:
    events = df_events.loc[df_events["trans_code"].eq(direction)]
    trades = df_sp_trades.loc[df_sp_trades["trans_code"].eq(direction)]
    sample_rows.append({"direction": label, "trades": len(trades), "events": len(events),
                        "companies": events["issuer_cik"].nunique(),
                        "first filing": trades["filing_date"].min().date(),
                        "last filing": trades["filing_date"].max().date(),
                        "first event": events["event_week"].min().date(),
                        "last event": events["event_week"].max().date(),
                        "all CARs available (%)": 100 * events[car_cols].notna().all(axis=1).mean()})
display(pd.DataFrame(sample_rows).set_index("direction").round(2))
coverage = df_events.groupby("trans_code")[car_cols].count().rename(index={"P": "Purchases", "S": "Sales"})
display(coverage)
typical = purchase_events[["total_value", "n_insiders", "beta", "mean_filing_lag"]].describe(percentiles=[.25,
.5, .75]).T
typical["IQR"] = typical["75%"] - typical["25%"]
typical.index = ["Purchase value ($)", "Distinct insiders (count)", "Beta (unitless)", "Mean filing lag
(calendar days)"]
display(typical.round(2))
role_counts = purchase_events["role"].value_counts().rename("events").to_frame()
role_counts["share (%)"] = 100 * role_counts["events"] / len(purchase_events)
display(role_counts.round(2))
plan_events = purchase_events.loc[purchase_events["first_filing"].ge("2023-01-01")]
plan_flags = plan_events["any_plan_10b5_1"]
display(pd.Series({"2023+ purchase events": len(plan_events), "observed flags": plan_flags.notna().sum(),
                   "missing flags": plan_flags.isna().sum(), "flagged events": plan_flags.eq(1).sum(),
                   "flagged among observed (%)": 100 * plan_flags.mean(),
                   "mixed purchase events": purchase_events["mixed_week"].sum(),
                   "mixed purchase events (%)": 100 * purchase_events["mixed_week"].mean()},
name="value").round(2))
```

	trades	events	companies	first filing	last filing	first event	last event	all CARs available (%)
direction								
Purchases	6358	2356	459	2020-01-02	2025-12-31	2020-01-03	2026-01-02	94.52
Sales	121010	25464	582	2020-01-02	2025-12-31	2020-01-03	2026-01-02	93.75

	car_1w	car_4w	car_13w	car_26w
trans_code				
Purchases	2227	2227	2227	2227
Sales	23873	23873	23873	23873

Table shown with fields as rows for print readability.

	Purchase value ($)	Distinct insiders (count)	Beta (unitless)	Mean filing lag (calendar days)
count	2356.0	2356.0	2227.0	2356.0
mean	10834728.406183	1.477929	1.019032	9.381185
std	104216477.702006	1.627243	0.511647	37.24998
min	0.4625	1.0	-0.568833	0.0
25%	72589.0152	1.0	0.669946	1.0
50%	241795.0	1.0	0.995258	2.0
75%	980135.0	1.0	1.311362	3.333333
max	3121679536.26	12.0	3.118219	365.0
IQR	907545.9848	0.0	0.641417	2.333333

	events	share (%)
role		
Director	1427	60.57
C-suite	563	23.90
Other officer	239	10.14
10% owner	117	4.97
Other	10	0.42

```
2023+ purchase events         1102.00
observed flags                 897.00
missing flags                  205.00
flagged events                  50.00
flagged among observed (%)       5.57
mixed purchase events          454.00
mixed purchase events (%)       19.27
Name: value, dtype: float64
```

Findings.

Purchases account for **2,356 of 27,820 events (8.47%)**. Of the purchase events, **2,227 (94.52%)** have returns at every horizon. Median purchase value is **$241,795**, with the middle half between **$72,589 and $980,135** (IQR **$907,546**). The median event has **one insider**, beta **1.00**, and mean filing lag **2 calendar days**.

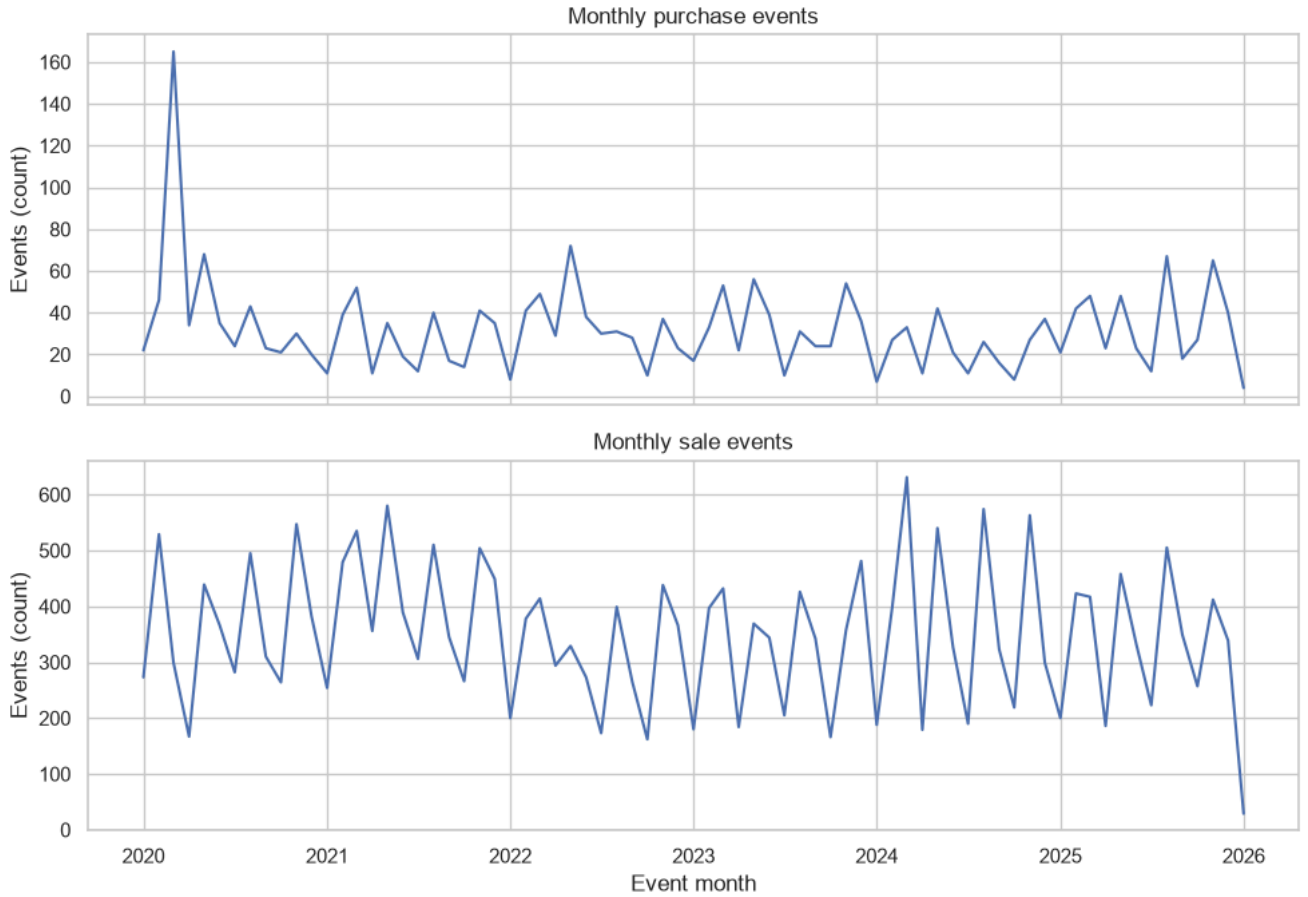
There are **454 mixed purchase events (19.27%)**, which supports a later sensitivity analysis excluding weeks with both directions. Among **1,102 purchase events first filed in 2023 or later**, **897** have recorded plan flags and **50 (5.57% of observed flags)** are flagged. Missing and earlier flags are not treated as evidence of no plan.

Event dates extend to **January 2, 2026** because the entry rule moves some late-2025 filings into the next Friday. All four horizons have the same available-event counts: the saved prices extend beyond the filing period, so this run does **not** show extra sample loss at 26 weeks.

5.2 Events Over Time

We next count events by month to see when buying and selling cluster. Separate panels keep the larger sale sample from hiding purchase variation. January 2026 is partial and contains events assigned from late-2025 filings.

```
# 5c. Plot monthly event counts with separate scales for purchases and sales.
event_month = df_events["event_week"].dt.to_period("M")
monthly_events = (df_events.groupby([event_month, "trans_code"]).size().unstack(fill_value=0)
                  .reindex(pd.period_range(event_month.min(), event_month.max(), freq="M"), fill_value=0))
fig, axes = plt.subplots(2, 1, figsize=(10, 7), sharex=True)
for ax, direction, label in zip(axes, ["P", "S"], ["Purchase", "Sale"]):
    sns.lineplot(x=monthly_events.index.to_timestamp(), y=monthly_events[direction].to_numpy(), ax=ax)
    ax.set(title=f"Monthly {label.lower()} events", ylabel="Events (count)", xlabel="Event month")
fig.tight_layout()
plt.show()
```



Findings.

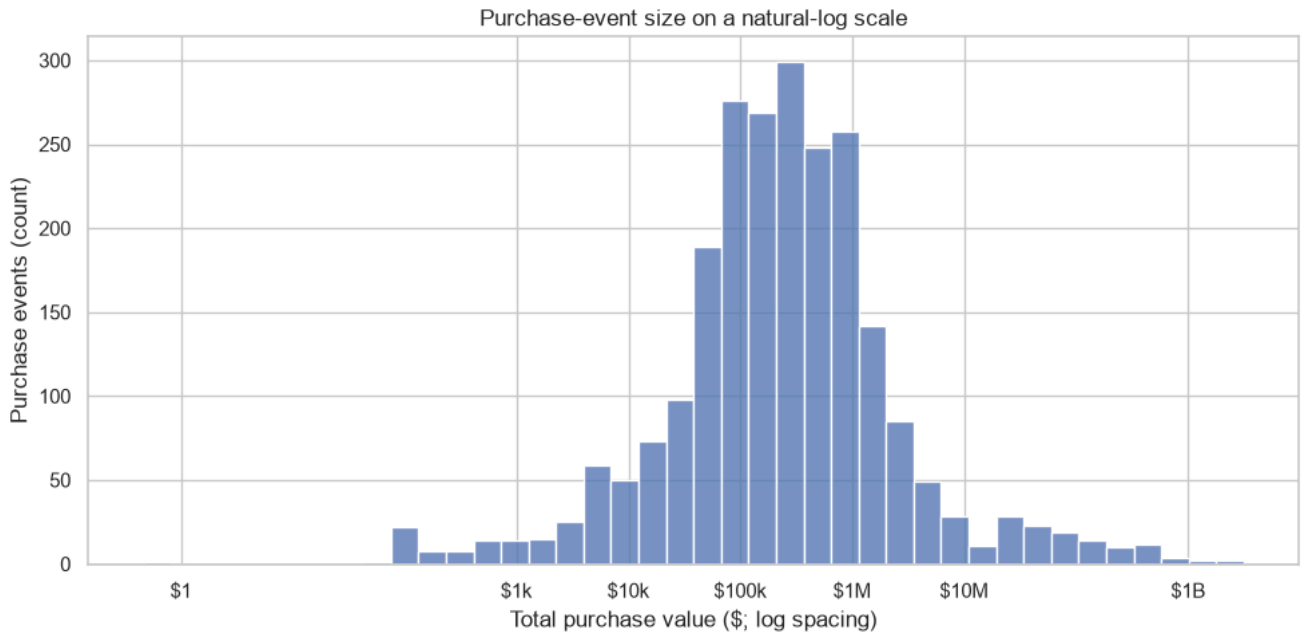
The highest purchase month is **March 2020 with 165 events**, compared with a median of **29** across observed purchase months. March 2020 clustering is consistent with purchases around the market downturn, but the count alone does not establish why insiders traded. Time clustering means observations

should not be interpreted as independent repetitions of identical market conditions.

5.3 Purchase Size

We then examine purchase amounts. The histogram uses `log_value` so small and large events can be shown together, with dollar labels for interpretation. This checks whether the planned log transformation is appropriate; all positive values are retained.

```
# 5d. Plot log purchase values with dollar labels on the logarithmic coordinates.
fig, ax = plt.subplots(figsize=(10, 5))
sns.histplot(data=purchase_events, x="log_value", bins=40, ax=ax)
dollar_ticks = np.array([1, 1_000, 10_000, 100_000, 1_000_000, 10_000_000, 1_000_000_000])
ax.set_xticks(np.log(dollar_ticks), ["$1", "$1k", "$10k", "$100k", "$1M", "$10M", "$1B"])
ax.set(title="Purchase-event size on a natural-log scale", xlabel="Total purchase value ($; log spacing)",
ylabel="Purchase events (count)")
fig.tight_layout()
plt.show()
```



Findings.

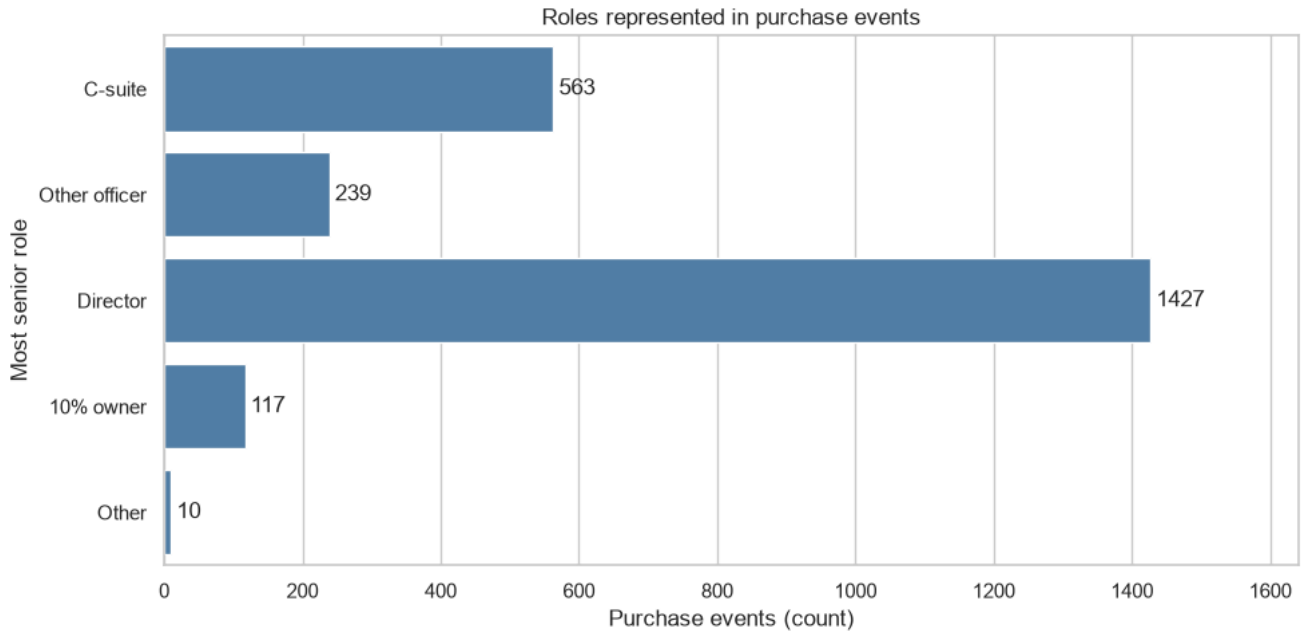
The median is **$241,795**, while the mean is **$10,834,728**, showing that large events pull up the average. Values range from **$0.46 to $3,121,679,536**. The log transformation compresses this range without deleting extremes. Very small and very large recorded events deserve review before Phase 3.

5.4 Insider Roles

We next count purchase events by their highest-ranked insider role to check the size of each group for the proposed role analysis. Each bar counts events, not individual people.

```
# 5e. Count purchase events by their most senior recorded insider role.
fig, ax = plt.subplots(figsize=(10, 5))
role_order = ["C-suite", "Other officer", "Director", "10% owner", "Other"]
```

```
sns.countplot(data=purchase_events, y="role", order=role_order, color="steelblue", ax=ax)
for container in ax.containers:
    ax.bar_label(container, padding=3)
ax.set(title="Roles represented in purchase events", xlabel="Purchase events (count)", ylabel="Most senior
role")
ax.set_xlim(0, purchase_events["role"].value_counts().max() * 1.15)
fig.tight_layout()
plt.show()
```



Findings.

Directors account for **1,427 events (60.6%)**, followed by **563 C-suite**, **239 other-officer**, **117 10%-owner**, and **10 other events**. C-suite events can be contrasted with the remaining **1,793 events** in the planned model, subject to return availability. The small Other category does not support a detailed standalone comparison.

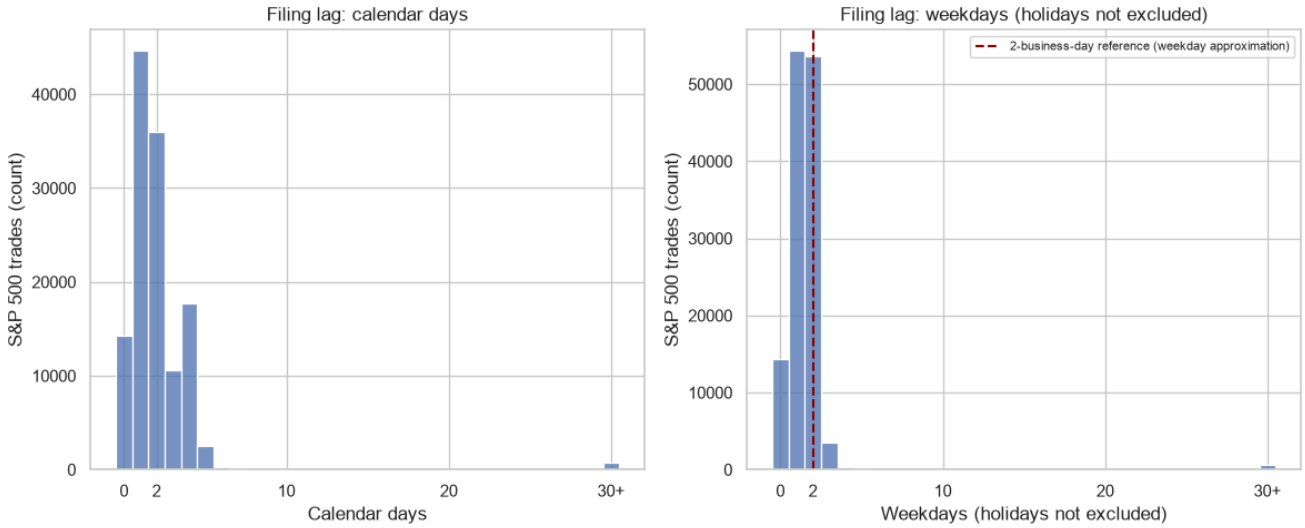
5.5 Filing Delays

We examine the gap between transaction and filing dates because it affects when information becomes available. The left panel shows calendar days; the right excludes weekends but not holidays. The two-business-day reference is shown on the weekday panel. Lags above 30 are included in the 30+ bin; this chart does not classify legal compliance.

```
# 5f. Distinguish calendar-day filing lags from weekday lags and retain the long tail in a 30+ bin.
weekday_lag = np.busday_count(df_sp_trades["trans_date"].to_numpy(dtype="datetime64[D]"),
                              df_sp_trades["filing_date"].to_numpy(dtype="datetime64[D]"))
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
for ax, values, label in [(axes[0], df_sp_trades["filing_lag_days"], "Calendar days"),
                          (axes[1], weekday_lag, "Weekdays (holidays not excluded)")]:
    sns.histplot(x=np.minimum(values, 30), bins=np.arange(-.5, 31, 1), ax=ax)
    ax.set(title=f"Filing lag: {label.lower()}", xlabel=label, ylabel="S&P 500 trades (count)")
    ax.set_xticks([0, 2, 10, 20, 30], ["0", "2", "10", "20", "30+"])
axes[1].axvline(2, color="darkred", linestyle="--", label="2-business-day reference (weekday approximation)")
axes[1].legend(fontsize=8)
```

```
print("Calendar lag percentiles:", df_sp_trades["filing_lag_days"].quantile([.5, .9, .99]).to_dict())
print("Trades beyond 30 calendar days:", df_sp_trades["filing_lag_days"].gt(30).sum())
fig.tight_layout()
plt.show()
```

```
Calendar lag percentiles: {0.5: 2.0, 0.9: 4.0, 0.99: 6.0}
Trades beyond 30 calendar days: 710
```



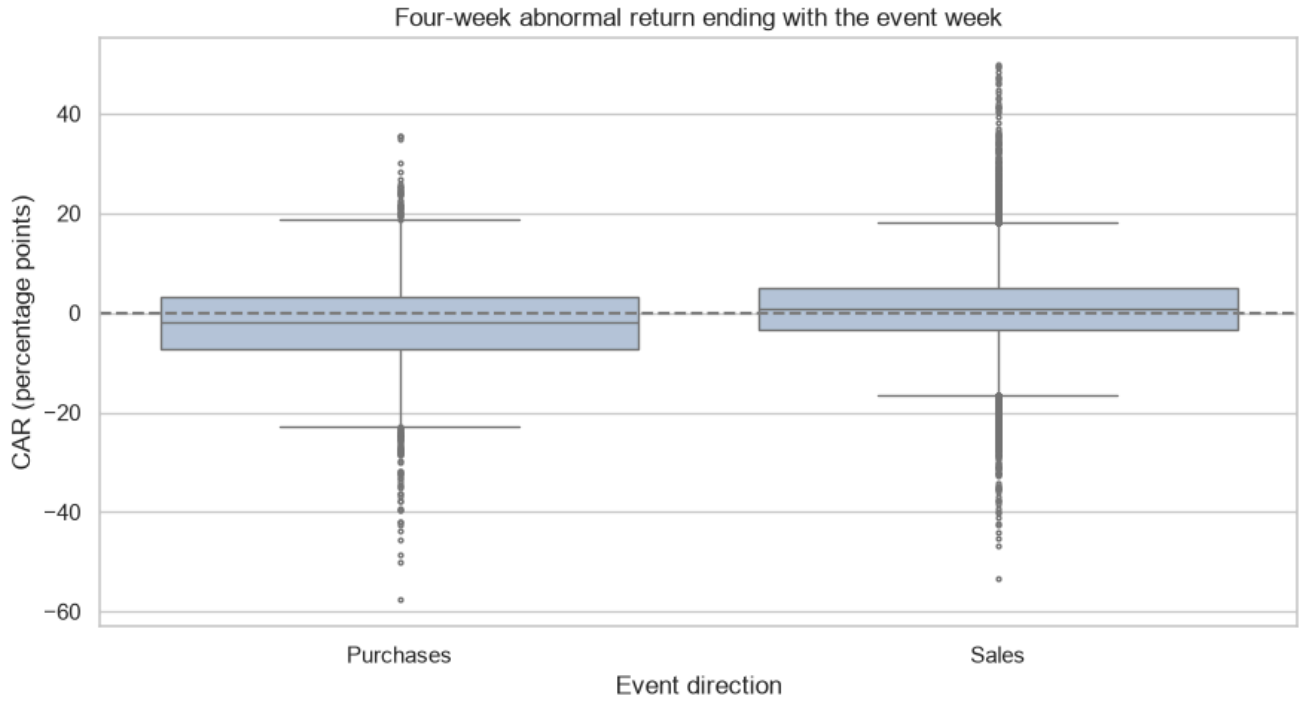
Findings.

For the **S&P 500 trade sample**, median lag is **2 calendar days**, the 90th percentile is **4**, and the 99th percentile is **6**. There are **710 trades (0.56%)** beyond 30 days. Timing variation supports anchoring events to disclosure rather than transaction dates. The weekday approximation cannot identify deadline violations, and the event-date rule itself still requires holiday-aware review before stronger timing claims.

5.6 Recent Price Movement

We then examine `car_pre4` to see whether purchases and sales occur around different recent price movements. The four-week window ends with the event week, so it can include a reaction to the disclosure. Boxplots show the median, middle half, and tails, retaining outlier points.

```
# 5g. Compare the four-week diagnostic ending with the event week across purchase and sale events.
pre_plot = df_events[["trans_code", "car_pre4"]].dropna().copy()
pre_plot["Direction"] = pre_plot["trans_code"].map({"P": "Purchases", "S": "Sales"})
pre_plot["CAR (percentage points)"] = pre_plot["car_pre4"] * 100
fig, ax = plt.subplots(figsize=(9, 5))
sns.boxplot(data=pre_plot, x="Direction", y="CAR (percentage points)", order=["Purchases", "Sales"],
            color="lightsteelblue", fliersize=2, ax=ax)
ax.axhline(0, color="gray", linestyle="--")
ax.set(title="Four-week abnormal return ending with the event week", xlabel="Event direction")
fig.tight_layout()
plt.show()
```



Findings.

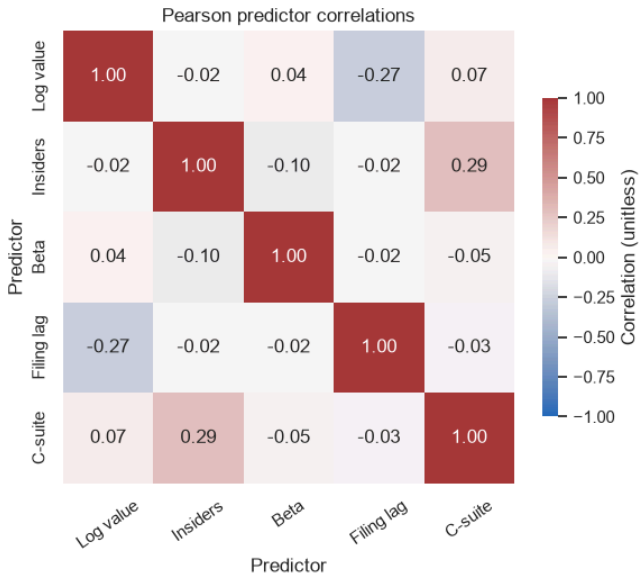
The median diagnostic is **-1.80 percentage points for purchases** and **0.81 for sales**. This is consistent with buying around recent declines and selling around recent gains. It does not show that insiders anticipated subsequent performance or establish a causal explanation. The inclusion of the event week limits a strictly pre-disclosure interpretation.

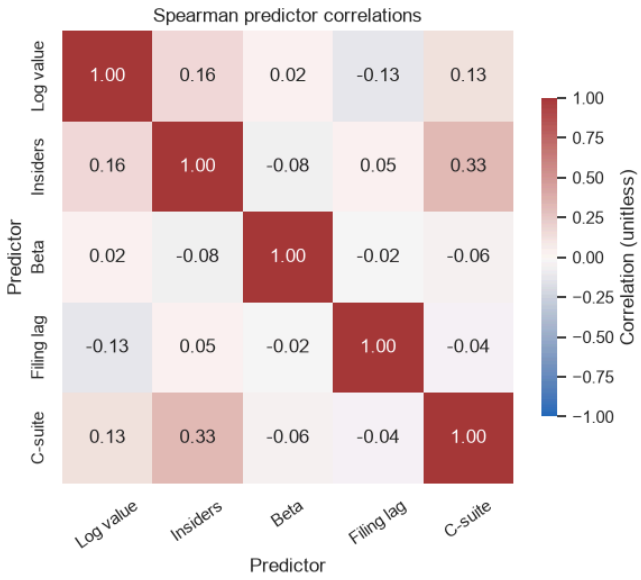
5.7 Relationships Between Predictors

Finally, we check whether the proposed predictors carry similar information. Pearson correlations compare their numerical values, and Spearman correlations compare their ranks. Both matrices use purchase events with all five predictors observed. They include no post-event return outcome.

```
# 5h. Inspect Pearson and Spearman correlations among predictors only on one complete-case sample.
predictor_cols = ["log_value", "n_insiders", "beta", "mean_filing_lag", "any_c_suite"]
predictors = purchase_events[predictor_cols].dropna()
predictor_labels = ["Log value", "Insiders", "Beta", "Filing lag", "C-suite"]
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
for ax, method in zip(axes, ["pearson", "spearman"]):
    correlation = predictors.corr(method=method)
    sns.heatmap(correlation, annot=True, fmt=".2f", vmin=-1, vmax=1, center=0, cmap="vlag", square=True,
                xticklabels=predictor_labels, yticklabels=predictor_labels,
                cbar_kws={"label": "Correlation (unitless)", "shrink": .7}, ax=ax)
    ax.set(title=f"{method.title()} predictor correlations", xlabel="Predictor", ylabel="Predictor")
    ax.tick_params(axis="x", rotation=35)
print(f"Complete purchase events: {len(predictors):,}; excluded for missing predictors: {len(purchase_events)
- len(predictors):,}")
fig.tight_layout()
plt.show()
```

```
Complete purchase events: 2,227; excluded for missing predictors: 129
```





Findings.

The matrices use **2,227 purchase events**, excluding **129** with incomplete predictors. The largest absolute off-diagonal correlation is **0.29 for Pearson** and **0.33 for Spearman**. C-suite participation and insider count have Pearson correlation **0.29**. These pairwise checks do not show near-duplicate predictors, but they cannot rule out multivariable collinearity; VIF and model diagnostics remain part of the later proposed analysis.

5.8 Returns After Disclosure

Having examined the sample and its predictors, we summarize purchase and sale CARs separately at each horizon. We report the mean, median, standard deviation, and 5th and 95th percentiles in **percentage points**. These describe the outcomes without fitting either planned research model or comparing returns by role or purchase size.

```
# 5b. Summarize each direction's outcome distribution separately without fitting a research model.
for label, events in [("Purchase events", purchase_events), ("Sale events", sale_events)]:
    print(f"{label}: CAR in percentage points")
    distribution = (events[car_cols] * 100).describe(percentiles=[.05, .5, .95]).T
    display(distribution[["count", "mean", "50%", "std", "5%", "95%"]]
            .rename(columns={"50%": "median"}).round(2))
```

```
Purchase events: CAR in percentage points
```

	count	mean	median	std	5%	95%
car_1w	2227.0	0.02	0.04	4.72	-7.07	6.86
car_4w	2227.0	0.33	0.31	8.59	-12.86	13.85
car_13w	2227.0	2.13	1.75	17.30	-24.45	27.81
car_26w	2227.0	3.13	2.39	26.63	-38.44	45.90

```
Sale events: CAR in percentage points
```

	count	mean	median	std	5%	95%
car_1w	23873.0	-0.18	-0.17	3.73	-5.88	5.47
car_4w	23873.0	-0.90	-0.79	7.83	-13.09	11.06
car_13w	23873.0	-2.47	-2.31	15.21	-26.25	21.12
car_26w	23873.0	-5.22	-4.87	23.52	-41.92	30.28

Purchase events.

At 13 weeks, the purchase distribution has mean **2.13 percentage points**, median **1.75**, and standard deviation **17.30**. Its 5th and 95th percentiles are **-24.45** and **27.81**.

Sale events.

At 13 weeks, the sale distribution has mean **-2.47 percentage points**, median **-2.31**, and standard deviation **15.21**. Its 5th and 95th percentiles are **-26.25** and **21.12**.

5.9 Findings and Next Steps

- **Purchases are uncommon:** 2,356 of 27,820 events (8.47%). We keep purchases as the main focus and sales as the comparison group.
- **Purchases cluster in time:** March 2020 has 165 events, the highest monthly count. Common market conditions need to be considered when interpreting the results.
- **Purchase values vary widely:** the median is $241,795, and the middle half spans $72,589 to $980,135. This supports using log value in the planned model.
- **The broad role comparison is feasible:** there are 563 C-suite purchase events before return exclusions, while the Other group has only 10. We retain the broad C-suite comparison rather than separate analyses of every small category.
- **Coverage limits the scope:** 2,227 purchase events have returns at every horizon, with no additional loss at 26 weeks. Only 50 of 897 observed later-period purchase flags indicate a plan, so the plan question remains a limited extension.
- **Timing and mixed weeks matter:** median filing lag is two calendar days; 19.27% of purchase events share a week with sales. We therefore use disclosure-based entry and plan a check excluding mixed weeks.

These findings retain the main research question while supporting the log-value predictor, broad role comparison, and later-period restriction for plan flags. The next sections explain the remaining limitations and specify the two Phase 3 models. Neither research model nor a post-event return comparison by role or purchase size has been run.

6. Limitations

Limits of the Data and Analysis

1. **Missing prices and selection bias.** Yahoo did not return 75 requested symbols. After ticker matching, 53 of 593 companies lack usable prices, affecting 4,511 retained trade records, including 271 purchases. At the event level, 1,665 events lack a price ticker and 55 more lack beta. Acquired, delisted, and failed companies are disproportionately difficult to retrieve, although missingness is not itself proof of failure. observed returns may overrepresent surviving firms; the direction of bias is not known, and results cannot be generalized to all originally eligible events.
2. **Large-company scope and imperfect membership matching.** The sample covers S&P 500 members at filing dates, not the whole market. Lakonishok and Lee (2001) report stronger predictability in smaller firms, so a large-company sample may have weaker signals. Community membership dates, the 10% ticker-CIK matching threshold, renamed tickers, and share classes can introduce selection or linkage errors. findings concern this matched large-company sample, and should not be extrapolated to small firms or treated as exact historical index reconstruction without additional checks.
3. **Weekly granularity and assumed entry timing.** Filing date plus `BDay(1)` counts weekdays, not exchange or SEC holidays, and does not use filing acceptance timestamps. Friday labels may represent Thursday prices on market holidays. The next-Friday rule can delay entry after a disclosure. measured returns depend on this timing convention, especially at one week; a guarantee that every entry price follows actual public availability requires a holiday-aware and timestamp-aware audit. `car_pre4` includes the event week and may include post-disclosure reactions.
4. **Overlapping events and shared market conditions.** Repeated company events can reuse the same subsequent weekly returns. In addition, 454 purchase events (19.27%) share a week with sales. March 2020 has 165 purchase events. rows are not independent, and raw event counts overstate independent information. Removing mixed weeks addresses conflicting signals, but does not remove all overlapping return windows or common shocks.
5. **Confounding company news.** Earnings announcements, buybacks, mergers, and changing fundamentals can influence both insider activity and stock returns. even a positive association cannot establish that the disclosure caused the return or that an insider had private information.
6. **Market-model assumptions and changing risk.** A single SPY factor cannot capture industry or size effects. Historical alpha and beta can change, particularly around volatile periods such as 2020. The implementation uses up to 52 available return rows, which need not represent 52 consecutive calendar weeks if a history has gaps; the forward-window endpoint check also permits a one-week buffer. some measured abnormal return can be model error or irregular price coverage. A near-zero or positive CAR is conditional on this benchmark, not an objective measure of mispricing.
7. **Sales and transaction-code interpretation.** Insiders can sell for liquidity, taxes, or diversification rather than pessimism. SEC P/S codes can include private transactions as well as exchange trades, so our operational P/S filter is broader than a verified exchange-only sample (SEC Form 4 instructions). sales are an imperfect comparison group, and the project's open-market wording should be interpreted as the P/S operational definition until individual transaction details are reviewed.
8. **Plan flags have limited coverage.** There are 17,060 missing event plan flags. Among 1,102 purchase events first filed in 2023 or later, 897 have observed flags, 205 have missing flags, and 50 have a recorded positive flag; the requirement began April 1, 2023. The event maximum can be zero even when some component flags are missing. missing or zero flags are not comprehensive proof of discretionary trading, and a plan comparison must be restricted to the later period with explicit observed-flag denominators. The flag does not certify legal compliance.

9. **Role and owner measurement.** Titles are self-reported. The C-suite rule recognizes CEO, CFO, COO, and non-Vice-President President titles among officers, which can miss other senior titles or assign ambiguous titles incorrectly. Filing-level roles select the highest-ranked reporting owner, while event insider counts use retained owner CIKs. role coefficients would describe our classification rather than a perfectly measured executive hierarchy, and joint filers can distort the link between a role and a particular trade.
10. **Amendments, duplicates, and recorded value extremes.** Amendments and cross-CIK duplicate rules are reproducible heuristics, not a manual audit of each filing. Purchase-event values range from $0.46 to about $3.12 billion. Large beneficial-owner filings may represent different behavior from ordinary executive purchases. residual reporting errors and unusual transactions can influence summaries and regression fits; log value reduces scale differences but does not establish validity. Any later exclusions must be documented rather than chosen to improve a result.
11. **Vendor prices and strategy realism.** Yahoo adjusted prices are a vendor product, not CRSP; adjustments, stale observations, share classes, and historical revisions may differ across providers. CAR adds abnormal weekly returns and omits transaction costs, taxes, execution constraints, and portfolio weighting. results describe benchmark-relative event outcomes, not the net return an outside investor could necessarily earn.

Impact on People

Publicly reported trading is not evidence of illegal insider trading. Attaching allegations to named individuals could damage reputations and misrepresent routine financial decisions. We report company-event patterns and avoid ranking or accusing individuals. Public source records can still identify people; aggregating and linking them can make personal financial behavior easier to scrutinize even when no new private data are collected.

A positive result could encourage retail investors to copy trades while bearing timing, concentration, and loss risks that the study does not measure. A null result could also be overstated as proof that disclosures never matter. We will report limitations alongside results, distinguish association from causation, and avoid presenting the project as investment advice. Missing-company coverage and classification errors can affect which people and firms appear successful, making unqualified rankings especially misleading.

7. Preregistration Statement 1

Purchases and Subsequent Returns

Prior analysis: Both plans were written after the EDA showed separate purchase and sale return distributions, but before fitting the research models or comparing returns by role or purchase size. These are not outcome-blind plans. Any later changes will be identified in the final report.

Hypothesis: The average **13-week CAR after purchase events is positive and exceeds the average after sale events**. The primary horizon is 13 weeks; 1, 4, and 26 weeks provide secondary descriptive context.

Motivation: Lakonishok and Lee (2001), *Are Insider Trades Informative?* find purchases more informative than sales, with important firm-size differences. Jeng, Metrick, and Zeckhauser (2003), *Estimating the Returns to Insider Trading: A Performance-Evaluation Perspective* report abnormal performance for purchase portfolios but not comparable evidence for sale portfolios. These findings motivate a directional hypothesis; their samples, weighting, and transaction timing differ from our public-disclosure event design and do not guarantee the result here.

Sample

The unit is one **company-week-direction event**, equally weighted regardless of its dollar value. Eligible filings are retained Form 4/4-A P/S records from 2020-2025 at companies matched to S&P 500 membership on the filing date. P/S is the operational purchase/sale definition and can include private transactions. Entry uses the **first Friday close on or after filing date plus one business day**, as implemented by a weekday offset; holiday and filing-time limitations are documented above.

The market model estimates alpha and beta using up to **52 available weekly returns strictly before the event week**, requiring at least **40**. Weekly abnormal return is actual stock return minus the model prediction (alpha plus beta times SPY return), written as `actual - (alpha + beta * market_return)` . `car_13w` sums the next 13 weekly abnormal returns, excluding the event week. It is stored in decimal units.

Use events with observed `car_13w` and a P/S direction. The current eligible sample is **26,100 events: 2,227 purchases and 23,873 sales**. Exclude missing price histories, insufficient estimation history, and missing outcomes; do not impute zero returns. Keep `mixed_week = 1` events in the primary analysis. Do not trim or winsorize valid outcomes based on their value. Report the retained counts and exclusions.

Planned Analysis

Fit ordinary least squares with an intercept using `sklearn.linear_model.LinearRegression`:

```
car_13w = intercept + beta_purchase * is_purchase + error
```

`is_purchase` is 1 for P and 0 for S, so **sales are the reference group**. The intercept estimates average sale CAR; `beta_purchase` estimates the purchase-minus-sale difference. The implied purchase mean is `intercept + beta_purchase` . The hypothesis predicts **both** `beta_purchase > 0` **and** `intercept + beta_purchase > 0` . Both conditions are needed to support the hypothesis.

Report both group means, the coefficient and implied purchase mean in percentage points, and purchase-only mean CARs at 1, 4, 13, and 26 weeks with their available counts. A decimal coefficient of 0.01 represents a one-percentage-point difference. **We will assess coefficient direction and size, without p-values or statistical-significance claims.** In particular, repeated events and overlapping windows do not justify treating every row as independent evidence.

Sensitivity analysis: Repeat the same 13-week model after excluding `mixed_week = 1` events, reporting the new counts and coefficient alongside the primary result. This sensitivity check does not fix other window overlap. Report positive, zero, negative, and inconsistent results, regardless of whether they support the hypothesis. A null or opposite sign will not trigger a switch of the primary horizon. Further specifications will be labeled exploratory.

8. Preregistration Statement 2

Insider Role and Purchase Size

Hypothesis: Among purchase events, **C-suite participation and larger dollar purchase values are associated with higher 13-week CARs**, controlling for the number of recorded insiders.

Motivation: Executives may have broad access to company information, and committing more dollars may indicate stronger conviction, although wealth and liquidity can also explain purchase size. Seyhun (1986), *Insiders' Profits, Costs of Trading, and Market Efficiency* motivates examining differences in information by insider position. Cohen, Malloy, and Pomorski (2012), *Decoding Inside Information* distinguish routine trading from more informative trading. Their classification is not equivalent to our C-suite flag or 10b5-1 checkbox. The literature motivates heterogeneity, not certainty that our exact role proxy or larger nominal purchases will predict better outcomes. Jeng, Metrick, and Zeckhauser (2003) also find that top executives do not outperform other insiders in their setting, so the role hypothesis faces meaningful contrary evidence.

Sample and Model

Use **purchase events only** (`trans_code = P`) with observed `car_13w` , `any_c_suite` , `log_value` , and `n_insiders` , currently **2,227 events**. Apply Statement 1's filing scope, event unit, entry rule, market model, and return definition. Exclude missing required values without imputation and report exclusions. Keep valid extreme values and mixed weeks in the primary model. Each event receives equal weight.

`any_c_suite` equals 1 when an event includes a trade assigned C-suite rank: an officer title matching CEO, CFO, COO, their expanded names, or President excluding Vice President. Zero includes directors, other officers, 10% owners, and Other; it is **not a directors-only reference group**. `log_value` is the natural logarithm of total reported purchase dollars. `n_insiders` counts distinct retained owner CIKs. Joint-filing and title limitations apply.

Fit the following multivariable linear regression with an intercept using `LinearRegression` :

```
car_13w = intercept + beta_c_suite * any_c_suite + beta_log_value * log_value + beta_insiders * n_insiders + error
```

The directional predictions are **`beta_c_suite > 0` and `beta_log_value > 0`** . Report the two predictions separately; support for one does not imply support for both. The C-suite coefficient is an adjusted difference in decimal CAR at the same recorded log value and insider count. For a 1% larger purchase, the exact modeled difference is `beta_log_value * log(1.01)` , approximately `0.01 * beta_log_value` in decimal CAR, holding other predictors fixed. Multiply by 100 to express percentage points. These are associations, not causal effects. No interactions, extra controls, or outcome-driven variable selection are included in the primary specification.

Model Checks and Reporting

1. **Collinearity:** Report predictor Pearson/Spearman correlations and variance inflation factors (VIF). Compute each VIF as `1 / (1 - R_squared)` from regressing one predictor on the others with an intercept. Large values indicate overlapping predictor information. Discuss the diagnostics without automatically dropping a prespecified variable to improve its coefficient.

2. **Prediction evaluation:** Make one random **70% training / 30% test split**, using `train_test_split(test_size=0.30, random_state=2950)`. Fit the fixed specification on the training subset only. Evaluate once on the held-out test subset with **RMSE**, the square root of mean squared prediction error, and compare with a baseline that always predicts the training outcome mean. Report RMSE in percentage points and both sample sizes. Examine a training residual-versus-fitted plot for systematic patterns and unequal spread; do not tune the model on test outcomes.
3. **Association estimates:** Fit the same fixed specification on all eligible purchase events for descriptive coefficients, explicitly distinguishing that full-sample fit from the model used for held-out RMSE. Report every coefficient, its units, and the sign of each prespecified directional prediction. No t-tests, p-values, or statistical-significance claims are planned.
4. **Dependence caveat and sensitivity:** A random event split can place the same company and overlapping outcome periods in both subsets. Test RMSE therefore measures this random-split task and is not an estimate of independent future-period performance. Repeat the fixed full-sample association model after excluding mixed weeks, and present it beside the primary model. A time-ordered or company-grouped validation scheme is a reviewer question for Phase 3, not an undisclosed replacement chosen after results.

Later-period extension: Leave `any_plan_10b5_1` out of the main model. A separately labeled exploratory Phase 3 analysis may consider 2023-2025 filings with observed flags, starting no earlier than April 1, 2023 and checking component-flag coverage. The small positive-flag purchase sample limits precision.

We will report results regardless of whether they support the hypotheses and document any changes to the specified model.

9. Questions for Reviewers

Following our group discussion, we seek feedback on these choices before fitting the Phase 3 models.

1. **Entry timing:** Is our Friday entry after filing date plus one weekday adequate, especially at the one-week horizon, or should we use filing acceptance timestamps and an exchange-holiday calendar? Would daily returns materially improve the short-horizon question?
2. **Repeated events:** Should we keep only the first company event within each 26-week window, or retain all events and limit claims about independence? Such thinning removes shared windows within a company but not common market shocks. How should we weigh the loss of purchase events against less overlap?
3. **Market benchmark:** Is the 52-observation SPY market model sufficient for this large-company sample, or should Phase 3 include a course-compatible industry or size adjustment? Should we require strictly consecutive estimation weeks rather than the last available weekly rows?
4. **Missing-company coverage:** With 53 companies lacking usable prices (75 failed symbols before company matching), is documenting the selection bias enough for Phase 3, or should we seek another source for acquired and delisted firms before interpreting returns?
5. **Role definition:** Is grouping CEO, CFO, COO, and non-Vice-President President as C-suite appropriate? Should directors be an explicit separate dummy rather than part of the broad non-C-suite reference, given 1,427 director and 563 C-suite purchase events before return exclusions?
6. **Validation and prior exposure:** Is the required random 70/30 event split acceptable if accompanied by its overlap caveat, or should a time-ordered/company-grouped check be primary? Since EDA already showed separate outcome distributions, is our explicit disclosure of that exposure sufficient for the course preregistration requirement?
7. **Plan and transaction definitions:** For the small 2023+ positive-plan purchase sample, should we defer the extension until filing-level checkbox coverage is audited? Should P/S transactions be reviewed for private transactions before using the term open-market in the final interpretation?